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Critical Thinking

Metode Penyelesaian Masalah dan Pemodelan

Dr. Indriana Hidayah, S.T., M.T.



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What is Thinking?

Why doesn't SHE like me?

Why doesn't HE like me?



As you start asking questions and seek answers, you are **thinking**.

What is Thinking?

Thinking is one of the primary cognitive skills that distinguishes humans from other creatures.



The ability to think allows us to **understand** the world, make **better decisions**, and develop **innovations**.



Types of Thinking

Studying various types of thinking helps us understand **the best ways to process information**, solve problems, and make decisions in different aspects of life

1. **Critical Thinking**
2. **Creative Thinking**
3. **Analytical Thinking**
4. **Abstract Thinking**
5. **Concrete Thinking**
5. **Divergent / Lateral Thinking**
6. **Convergent / Vertical Thinking**



Types of Thinking

Critical thinking involves asking questions and probing the underlying validity and assumptions behind statements or ideas. Critical thinking is very concerned with investigating the origins of claims and how they come about, and whether they come from hearsay or misinformation.

Creative thinking is always engaged with producing something new: new ideas, new theories, new solutions.

Analytical thinking is about breaking ideas and solutions become component parts and creating methods for categorizing and examining those parts using evidence and logic.

Abstracts thinkers can pull together seemingly disparate ideas together, and they can also reveal the hidden meanings behind concepts.



Types of Thinking

Concrete thinking is also known as practical thinking. Concrete thinking doesn't get carried away with context, analysis, or brainstorming. Its goal is to find the simplest solution as quickly as possible.

Divergent thinking is method is to find as many different solutions as possible in order to spark new ideas. Sometimes you even stray far away from the original problem in order to look for inspiration. This type of thinking is therefore often connected with creative and abstract thinking.

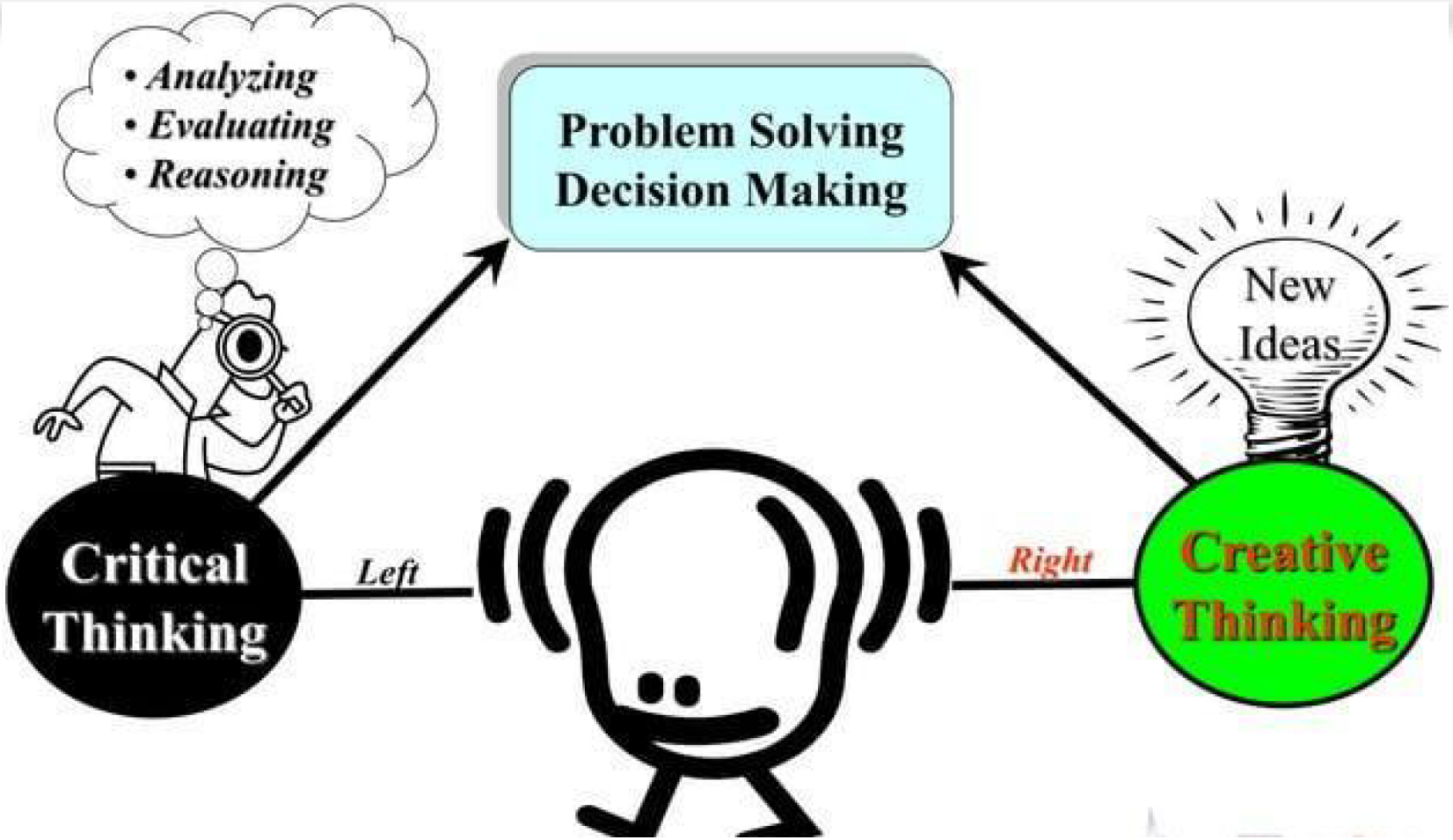
Convergent thinking is about finding one single path forward by combining ideas or eliminating them, through logic and analysis. Convergent thinking is very organized and focused on efficiency, but there is also some elegance and artfulness to it, since it does require combining and merging different ideas together.



Types of Thinking

Studying various types of thinking helps us understand **the best ways to process information**, solve problems, and make decisions in different aspects of life

Types of Thinking



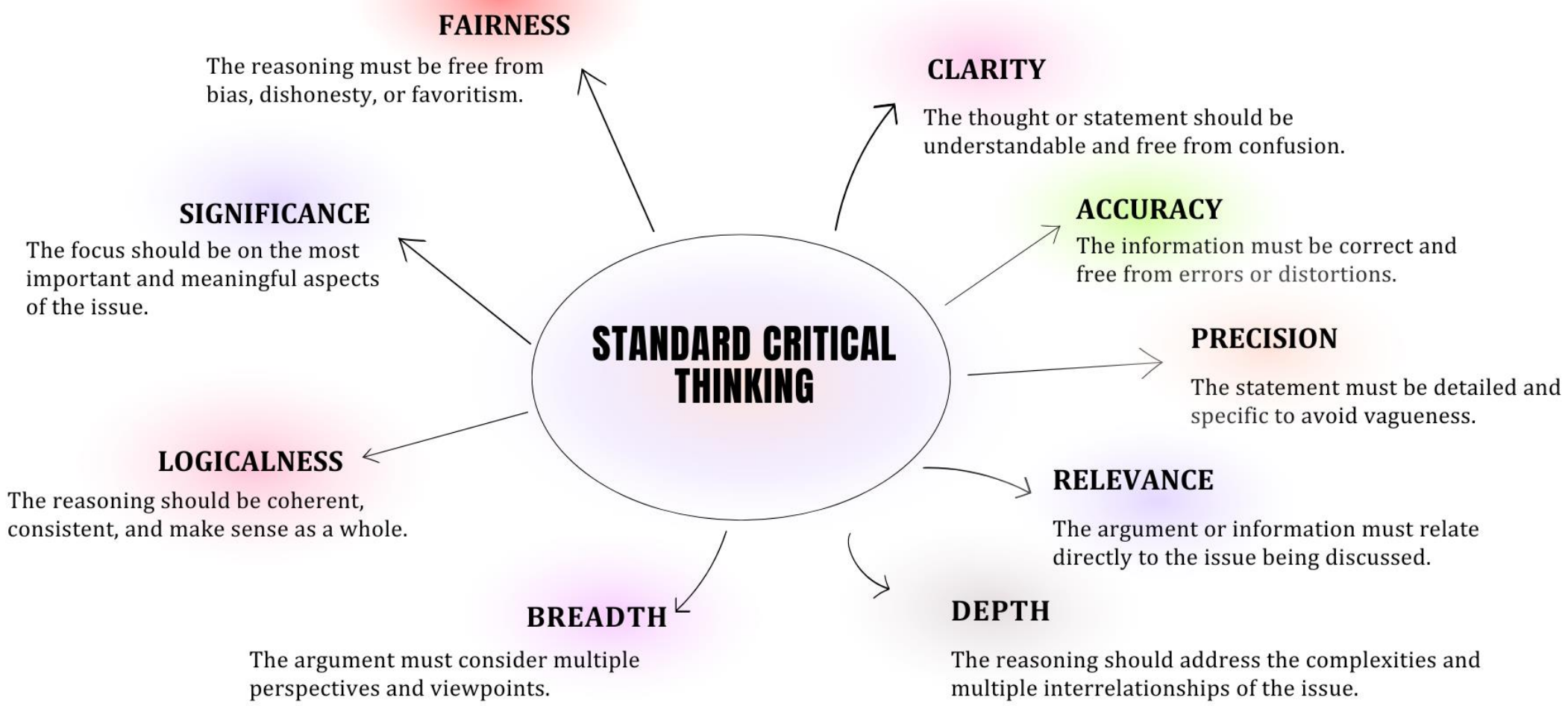


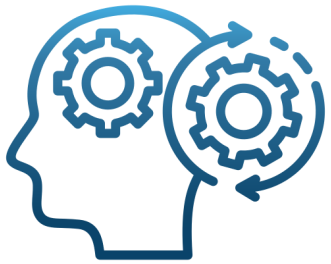
Critical Thinking

Critical Thinking is the general term given to a wide range of cognitive and intellectual skills needed to:

- Effectively identify, analyze, and evaluate arguments.
- Discover and overcome personal prejudices and biases.
- Formulate and present convincing reasons in support of conclusions.
- Make reasonable, intelligent decisions about what to believe and what to do.

Critical Thinking Standards (CTS)





CTS: clarity

- Could you elaborate further on that point?
- Could you express that point in another way?
- Could you give me an illustration?
- Could you give me an example?

CTS: accuracy

- Is that really true?
- How could we check that?
- How could we find out if that is true?



A statement can be clear but not accurate

CTS: precision

- Could you give more details?
- Could you be more specific?



A statement can be both clear and accurate, but not precise

CTS: precision

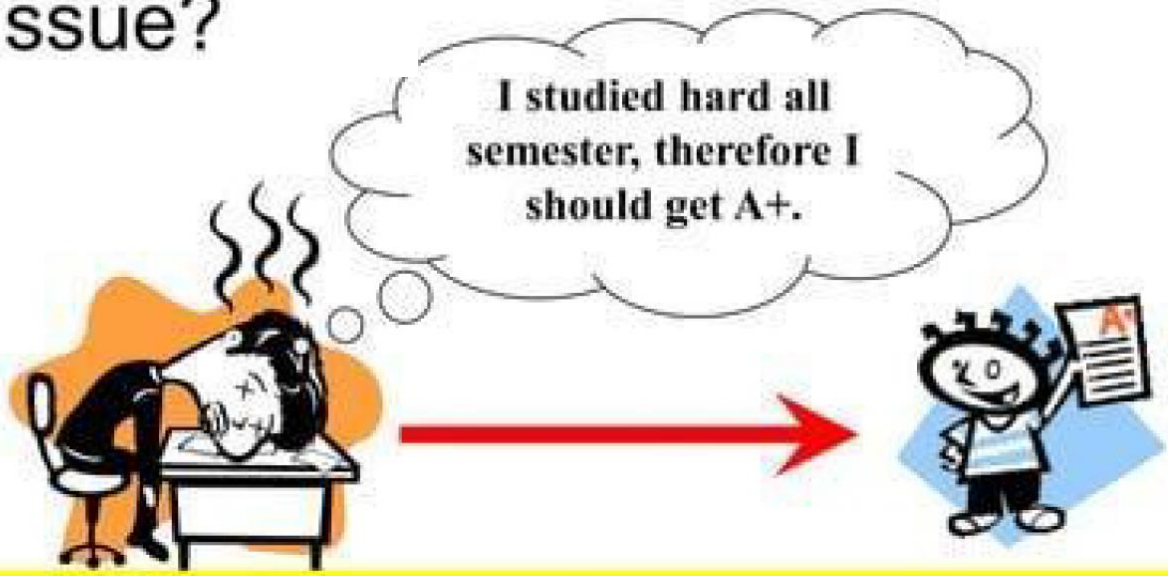
- Could you give more details?
- Could you be more specific?



A statement can be both clear and accurate, but not precise

CTS: relevance

- How is that connected to the question?
- How does that bear on the issue?

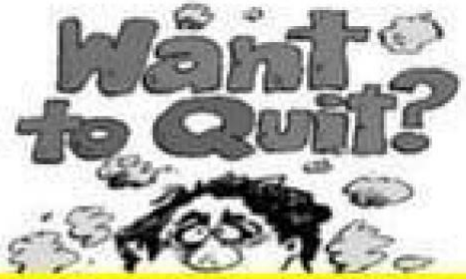


A statement can be clear, accurate, and precise, but not relevant to the question at issue.

CTS: depth

- How does your answer address the complexities in the question?
- How are you taking into account the problems in the question?
- Is that dealing with the most significant factors?

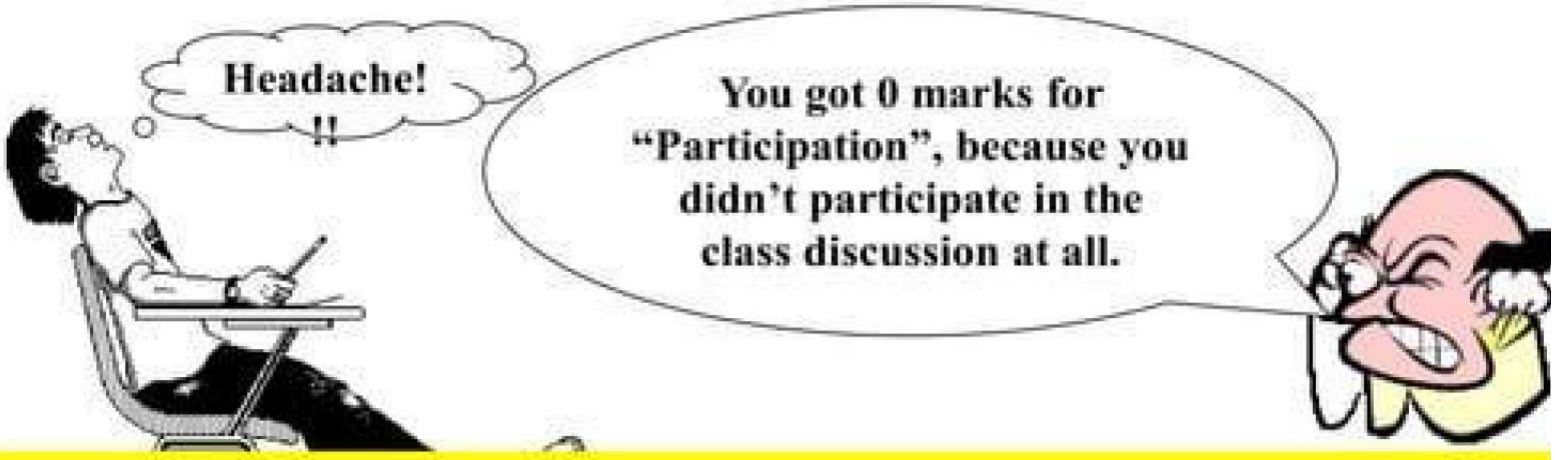




A statement can be clear, accurate, precise, and relevant, but superficial.

CTS: breadth

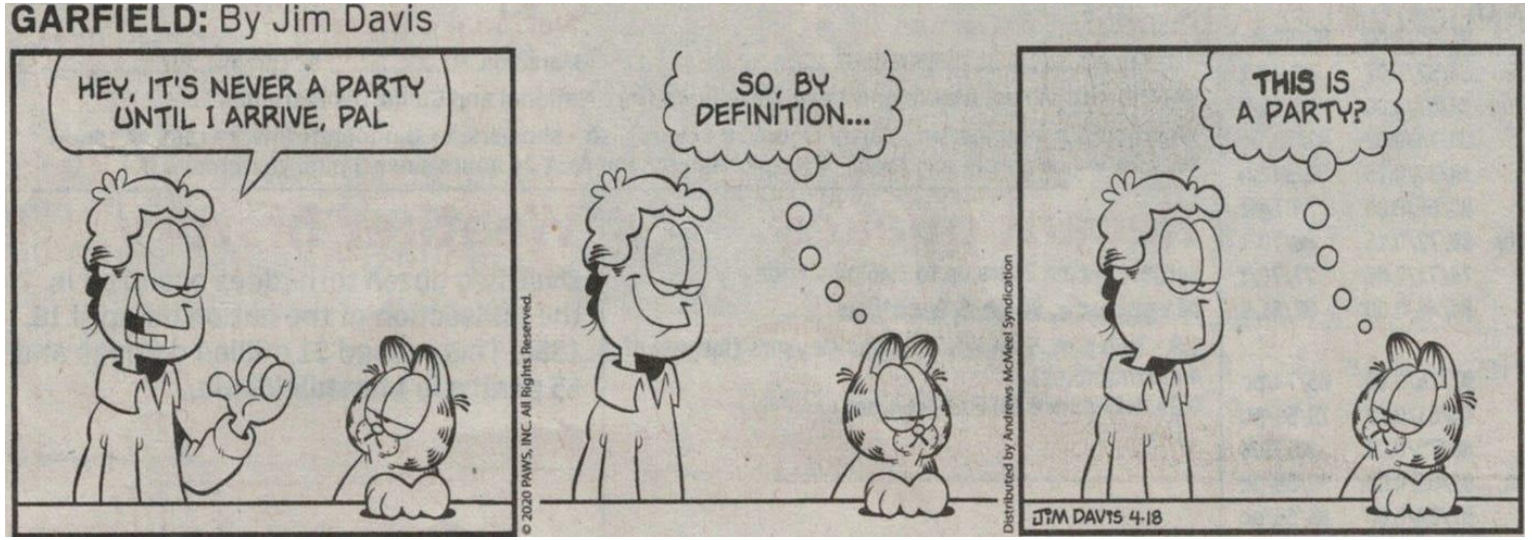
- Do we need to consider another point of view?
- Is there another way to look at this question?
- What would this look like from a conservative standpoint?
- What would this look like from the point of view of...?



A line of reasoning may be clear, accurate, precise, relevant, and deep, but lack breadth.

CTS: logic

- Does this really make sense?
- Does that follow from what you said?
- How does that follow?
- But before you implied this and now you are saying that; how can both be true?



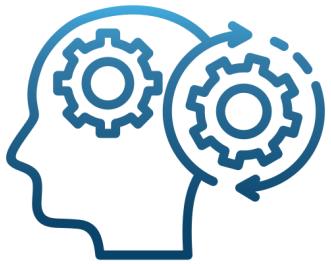
CTS: fair

Critical thinking demands that our thinking be fair.

- Open-minded
- Impartial
- Free of distorting biases and preconceptions



Fair-mindedness is an essential attribute of a Critical Thinker.



Generally, good thinking is…

CLEAR.....rather than.....UNCLEAR

ACCURATE.....rather than.....INACCURATE

PRECISE.....rather than.....VAGUE

RELEVANT.....rather than.....IRELEVANT

CONSISTENT....rather than.....INCONSISTENT

LOGICAL.....rather than.....ILLOGICAL

COMPLETE.....rather than.....INCOMPLETE

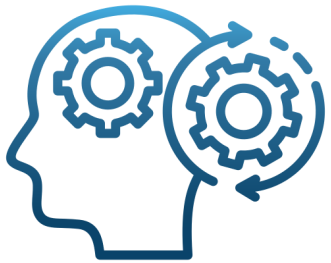
FAIR.....rather than.....BIASED



Benefits of Critical Thinking

Examples:

- **Academic Performance**
  - ❑ understand the arguments and beliefs of others
  - ❑ Critically evaluating those arguments and beliefs
  - ❑ Develop and defend one's own well-supported arguments and beliefs.
- **Workplace**
  - ❑ Helps us to reflect and get a deeper understanding of our own and others' decisions
  - ❑ Encourage open-mindedness to change
  - ❑ Aid us in being more analytical in solving problems
- **Daily life**
  - ❑ Helps us to avoid making foolish personal decisions.
  - ❑ Promotes an informed and concerned citizenry capable of making good decisions on important social, political and economic issues.
  - ❑ Aids in the development of autonomous thinkers capable of examining their assumptions, dogmas, and prejudices.



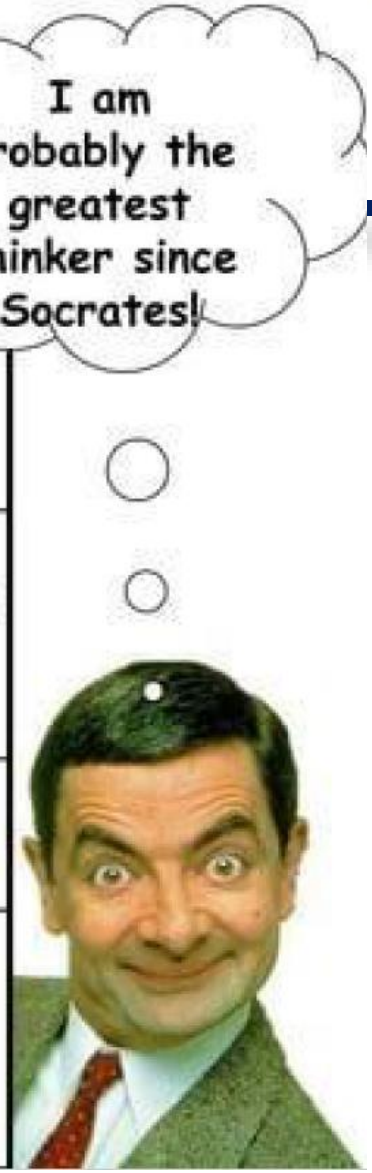
Barriers to Critical Thinking

If Critical Thinking is so important, why is it that ***uncritical thinking is so common***?

Why is that so many people including many highly educated and intelligent people ***find critical thinking so difficult?***

Five *Powerful Barriers* to Critical Thinking:

Egocentrism	**Self-centered thinking** • *self-interested thinking* • *self-serving bias*
Sociocentrism	**Group-centered thinking** • *Group bias* • *Conformism*
Unwarranted Assumptions	**Beliefs that are presumed to be true without adequate evidence or justification** • *Assumption* • *Stereotyping*
Wishful Thinking	**Believing that something is true because one wishes it were true.**
Relativistic Thinking	**The truth is "just a matter of opinion"** • *Relativism* ❑ *Subjectivism* ❑ *Cultural relativism*



Mini quiz

In a 1989 international study of 13-year-olds, Koreans finished first in mathematics and Americans finished last. Yet when asked whether they thought they were "good at mathematics," only 23 percent of Koreans said "yes," compared to 68 percent of Americans.

Which critical thinking barrier do the American students exhibit:

A) Self-interested thinking
B) Group bias
C) Self-serving bias
D) Conformism

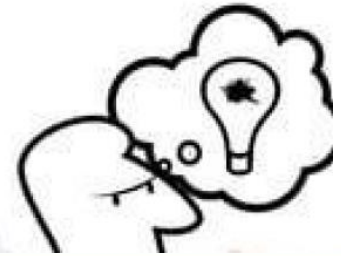


Mini quiz

Muhammad Ali [speaking in Zaire, Africa]: "There's no country as great as the smallest city in America. I mean [here in Zaire] you can't watch television. The water won't even run right. The toilets won't flush. The roads, the cars- there's nothing as great as America."

Which critical thinking barrier does Ali display in this passage?

A) Self-interested thinking
B) Group bias
C) Self-serving bias
D) Conformism



Mini quiz

Adam: My friend Andy is a 1st year student at UNITAR. He is cool, loves hanging out, and has a very laid-back personality.

Lee: I bet he's from KL.

Which critical thinking barrier does Lee exhibit?

A) Self-interested thinking
B) Stereotyping
C) Group bias
D) Conformism



Characteristics of critical thinker

Are you OPEN MINDED about other people's view?

Are you HONEST to yourself (or others) when you are wrong?

Do you have the COURAGE and PASSION to take initiative and confront problems and meet challenges?

Are you AWARE of your own biases and preconceptions?

Do you WELCOME CRITICISM from other peopl

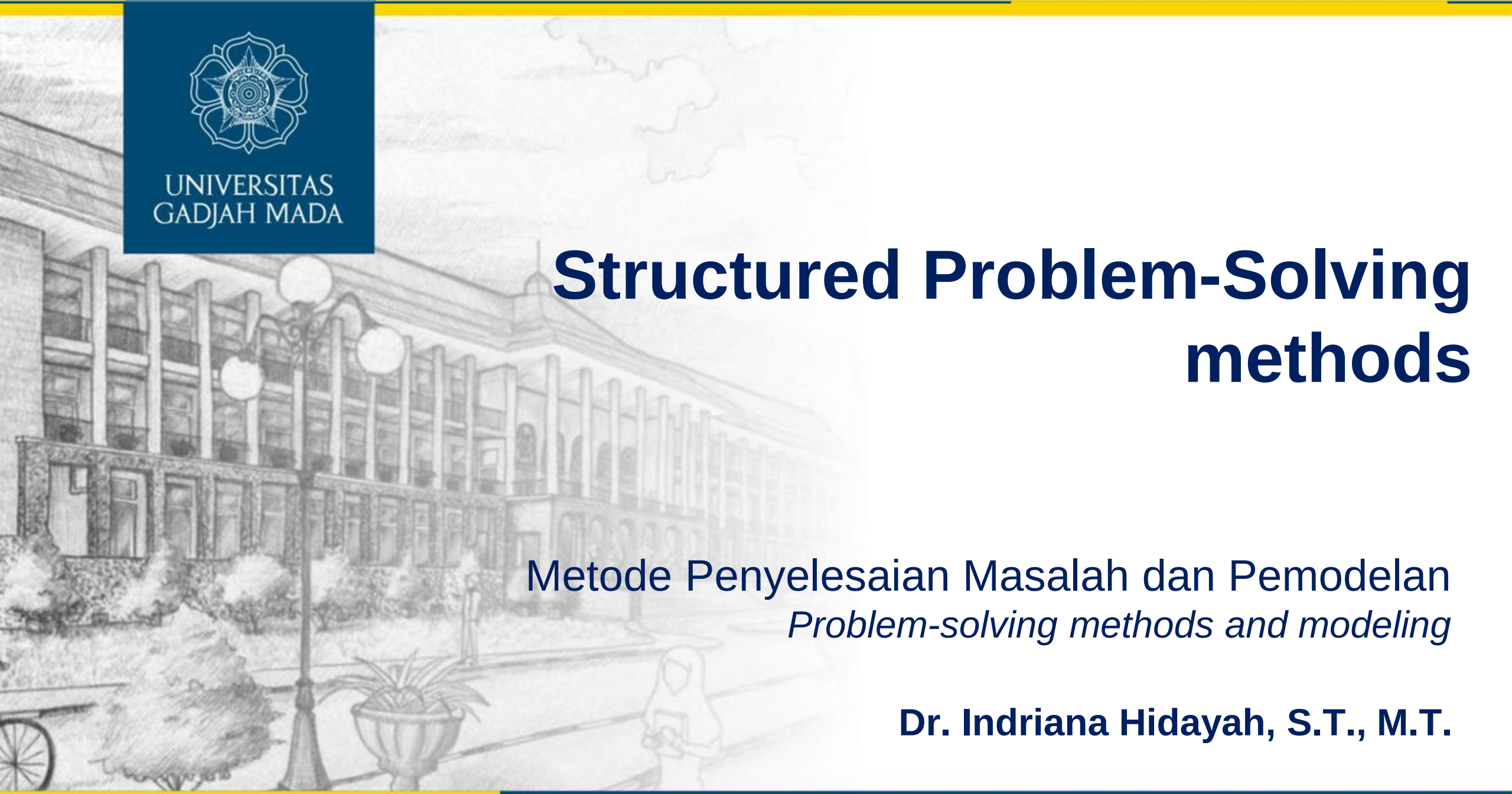
Do you have INDEPENDENT opinions and are not afraid to disa ee?



Critical thinking requires us to use our imagination, seeing things from perspectives other than our own and envisioning the likely consequences of our position

- Bell Hooks

Any questions?



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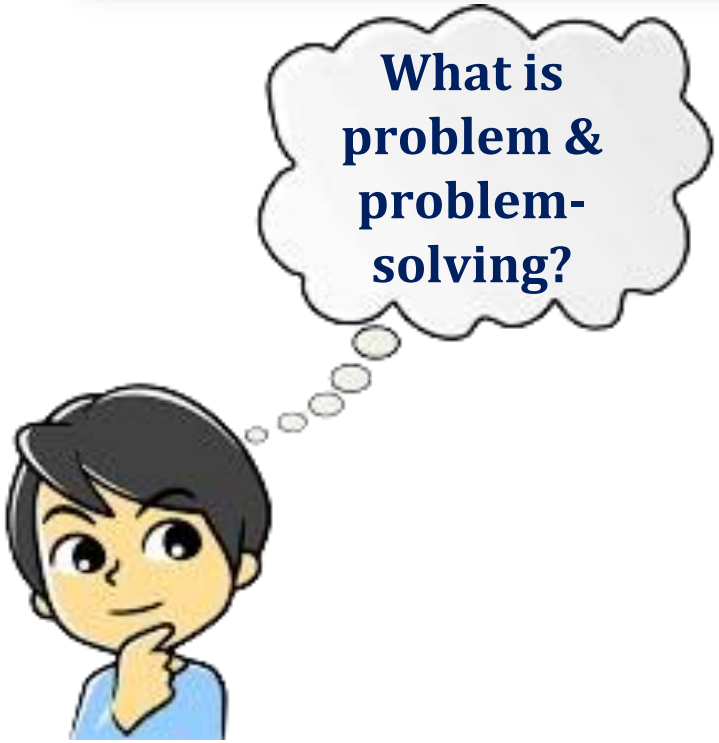
Structured Problem-Solving methods

Metode Penyelesaian Masalah dan Pemodelan

Problem-solving methods and modeling

Dr. Indriana Hidayah, S.T., M.T.

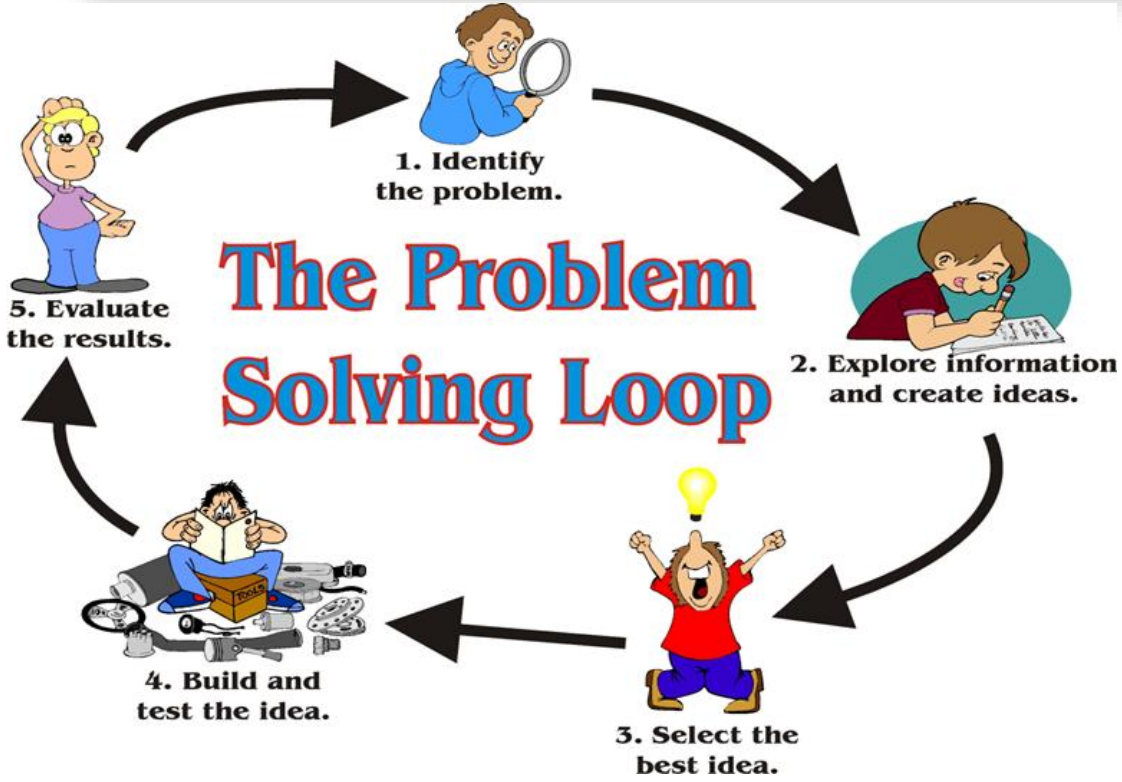
Problem-Solving



A **problem** is a situation, question, or obstacle that creates a gap between the current state and a desired outcome, requiring a solution or resolution. Problems arise when there is a discrepancy between expectations and reality, preventing progress or achieving goals.

Problem-solving is is the process of identifying a challenge or issue or problem and finding a solution.

General problem-solving steps



- One goal of this course is to train your problem-solving skill.

General problem-solving steps

- Identifying the Problem: Recognize and define the problem clearly.
- Gathering Information: Collect relevant data and information about the problem.
- Generating Solutions: Brainstorm possible solutions or approaches to address the problem.
- Evaluating Solutions: Assess the potential solutions and weigh their pros and cons.
- Choosing a Solution: Select the most appropriate and feasible solution.
- Implementing the Solution: Put the chosen solution into action.
- Reviewing the Results: Evaluate the outcome to see if the problem has been resolved effectively.

Identify problem: example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Clarify the problem**
    - **A tool called "is/is not diagram" can help in focusing on the true nature of the problem by eliminating assumptions and irrelevant details.**

Problem statement	Details
What it IS	Students need longer time to complete their study.
	…
What it IS NOT	Not exclusively cause by poor academic performance.
	…

Example

What it IS	Details
What it IS	- Graduate students taking longer than the expected duration to complete their studies.
	- Common across multiple departments and programs.
	- Occurs despite initial planning and timelines.
	- Leads to delays in graduation and entry into the job market.
	- Affects students' financial and personal plans.
What it IS NOT	- Not limited to a specific group of students (e.g., part-time vs. full-time).
	- Not exclusively caused by poor academic performance or failure.
	- Not a recent trend; it has been an ongoing issue.
	- Not unique to one university or institution.
	- Not related to the availability of resources (e.g., libraries, labs).

Identify problem: example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **"is/is not diagram" can also be used to help in this task.**

Possible contributing factors	IS a contributing factor	IS NOT a contributing factor
Academic workload	X	
Enrollment in required courses		X
…		

Example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **What it IS:**
      - Time Management: Graduate students may struggle with managing their time effectively.
      - Research Complexity: The complexity of research projects may lead to extended study periods.
      - Advisor Support: Variability in advisor availability and support can impact progress.
      - Funding: Limited or fluctuating funding can cause delays.
      - Life Events: Personal life events (e.g., family responsibilities, health issues) can extend study periods.

Example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **What it IS NOT:**
      - Academic Failure: The delays are not primarily due to academic failures or poor grades.
      - Resource Availability: It is not due to the lack of access to necessary resources like libraries or laboratories.
      - Limited to One Institution: This is not a problem unique to a single institution; it occurs across multiple universities.

Next Steps...?

- **Gather Data:**
  - **Collect data on time management practices,**
  - **research complexities,**
  - **advisor support,**
  - **funding stability, and**
  - **personal life events -> mental health**

Next Steps...?

- **Develop solution:**
  - **Generating Solutions: Brainstorm possible solutions**
  - **Evaluating Solutions: Assess the potential solutions and weigh their pros and cons.**
  - **Choosing a Solution: Select the most appropriate and feasible solution.**
- **Implementation**
- **Review the result**

Some structured Problem-Solving Methods

To be able to perform problem-solving, there are several methods that can be used, including:



- PDCA Cycle (Plan Do Check Act)
- Root Cause Analysis
- Define Measure Analyze Improve Control (DMAIC)
- 5 Whys Technique
- Theory of Inventive Problem Solving (TRIZ)

PDCA Cycle (Plan Do Check Act)

Definition

The PDCA cycle (Plan Do Check Act) is a methodology to develop and customize internal organizational procedures for quality, project, and problem management

The PDCA cycle (Plan Do Check Act) comprises four iterative steps:

- Plan (set objectives and develop a plan to meet a need)
- Do (implement the plan)
- Check (collect data and evaluate results)
- Act (take corrective action and standardize processes)





Step 1: Plan (PDCA)

The first step in the PDCA cycle (Plan) has four sub-steps:

1. **Identify problems**: The aim is to track down the root causes of problems.
   Example → If customers complain about product quality, it's crucial to identify the factors underlying the problem to eliminate it for good.

2. **Set objectives**: The organization's overall targets should serve as a focus for short- and medium-term objectives.
   Example → A manufacturing company might set itself the target of reducing product defects by 20% over the next six months. However, these objectives need to be SMART (specific, measurable, achievable, realistic, and timely) to see conclusive results.

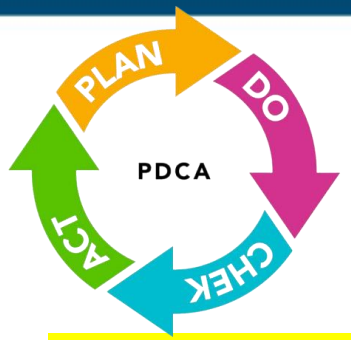


Step 1: Plan (PDCA)

3 **Develop a detailed plan:** Specify what needs to be accomplished, who is responsible for each task, how the implementation schedule will be phased, and what resources are required.
Example → A construction company looking to improve efficiency on its job sites, the detailed plan could include:

- Creating Gantt charts for each project
- Planning crew assignments
- Managing resources
- Purchasing new technologies to speed up construction processes

4 **Communicate the plan**: The final sub-step of the planning phase is sharing this plan with the whole organization. Everyone needs to understand their role and responsibilities in executing the PDCA framework before its implementation.



Step 2: Do (PDCA)

The Do phase of the PDCA process is used to transform planning into concrete action with solutions to the initial problem. It is made up of five sub-phases.

1. **Allocate the necessary resources:** Regarding workforce, materials, equipment, or budget, this ensures that the teams involved have the right resources to meet deadlines and quality standards.

2. **Train employees:** Training enables staff to acquire the skills to adapt to change, effectively use new tools, and contribute to successful implementation.

3. **Implement the plan:** Carry out the actions or tasks described methodically. For example, when introducing a new process in the manufacturing sector, it is essential to implement the plan and rigorously follow the steps to minimize potential losses.



Step 2: Do (PDCA)

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4 **Meet Deadlines:** It's imperative to meet the deadlines and milestones defined in the project plan to achieve the desired results. This discipline ensures that the project progresses according to schedule and avoids costly delays.

5 **Document:** Record all procedures and protocol details to meet regulatory requirements and enable in-depth performance analysis (equipment and materials used, unexpected problems, and deviations from the plan).



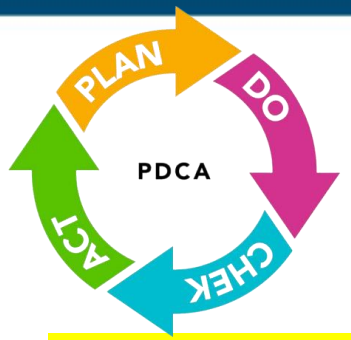
Step 3: Check (PDCA)

The Check step in the PDCA cycle is crucial for assessing the effectiveness of the actions taken. Gaps in organizational processes often become apparent at this stage. Therefore, it is essential to incorporate this step from the initial planning phase. The sub-steps include:

1. **Collect data:** Evaluate the effectiveness of the measures taken in the Do phase.
   Example: A manufacturing environment, collecting data on machine performance, downtime, and product quality is essential for assessing the efficiency of operations.

2. **Compare with predefined objectives**: Determining whether the plan has achieved the desired results is essential.
   Example → The distribution sector, comparing actual stock levels with the numbers defined at the planning stage helps to determine whether operations are on target.

3. **Analyze the root cause:** This is necessary in the event of deviation from objectives. When nonconformance to a clinical protocol occurs in the healthcare sector, it is essential to analyze the root causes, such as inadequate medical staff training or resource constraints, to ensure that patients receive quality care.



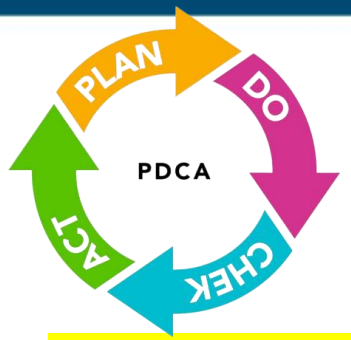
Step 3: Check (PDCA)

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4. **Share results:** Sharing results with all workers and managers promotes a culture of transparency and safety.

5. **Maintain or revisit the plan:** After assessing the performance of a strategy, you need to decide whether to keep, adjust, or review the approach if the results don't meet expectations.

6. **Document:** Future progress and permanent problem resolution depend on this sub-step. For example, in the healthcare field, detailed documentation of patient medical records and clinical procedures is necessary to ensure complete healthcare traceability and meet regulatory requirements.

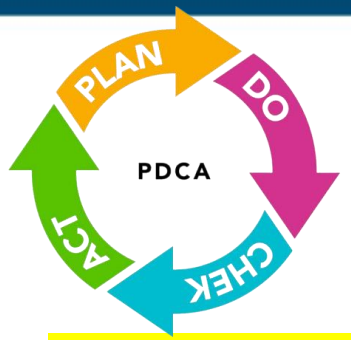


Step 4: Act (PDCA)

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1. **Adjust and reassess:** When results fall short of expectations, adjustments and reassessment are necessary.
   Example → If a company fails to meet its target of reducing product defects by 20% in six months, it can review its initial plan and make further changes.

2. **Provide feedback:** Feedback channels play a vital role in this phase. They enable the gathering of input from employees, customers, and other stakeholders.
   Example → Customer comments on product quality may indicate the need for further improvements.



Step 4: Act (PDCA)

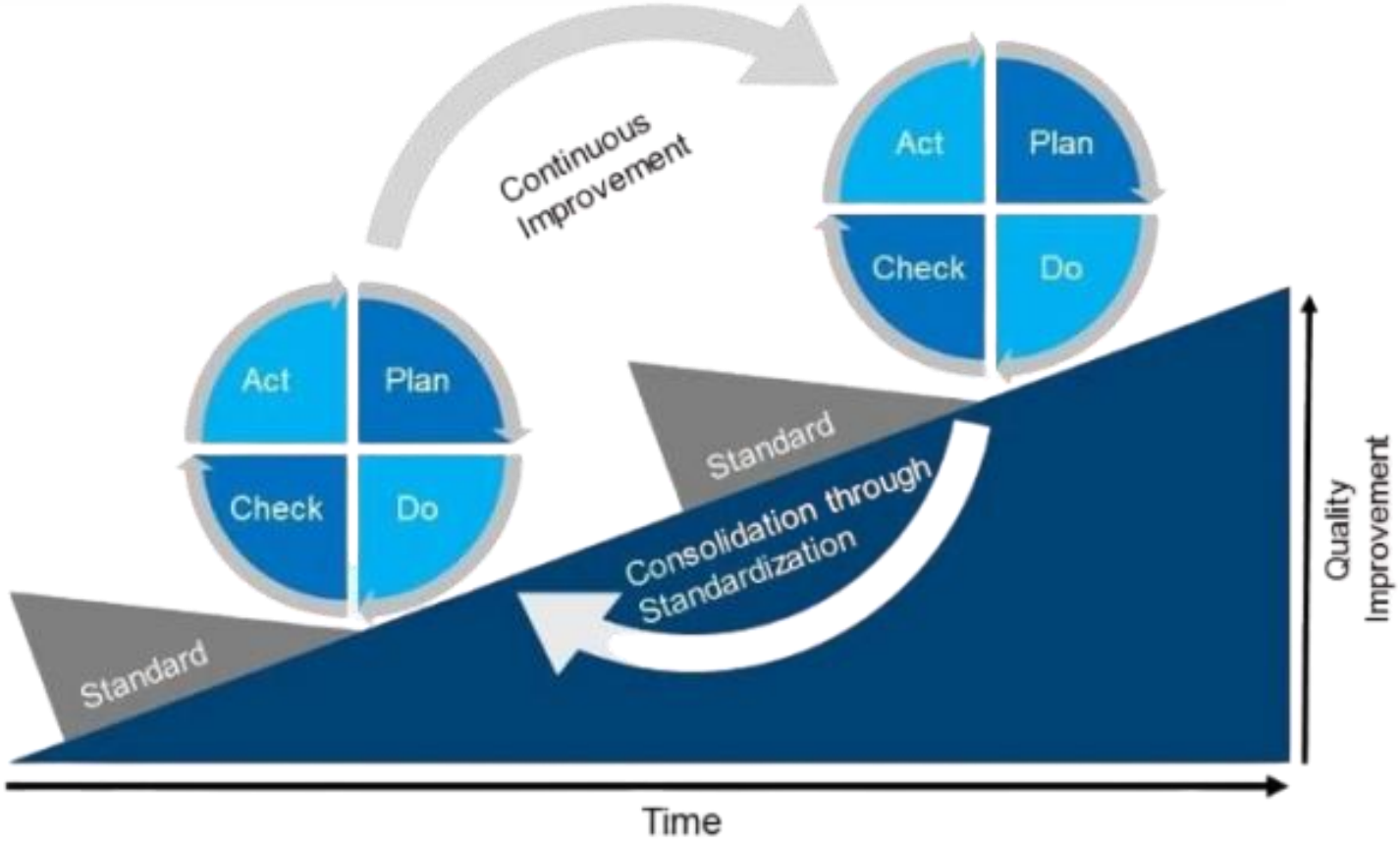
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3 **Standardize processes:** As part of the Act step, it is essential to deploy corrective measures and standardize all procedures. Standardization ensures that improvements are systematically integrated, and activities are carried out consistently. This process ensures operational stability, lasting success, and practical continuous improvement efforts.

4 **Promote a culture of continuous improvement:** This means emphasizing the development of a culture where the organization learns from experience, innovates, and adapts constantly.

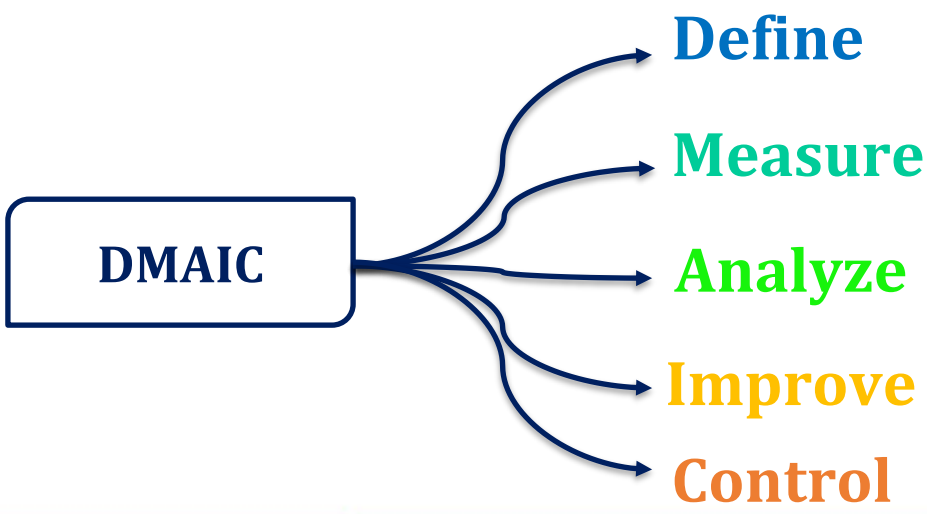
Iterative Loops of PDCA

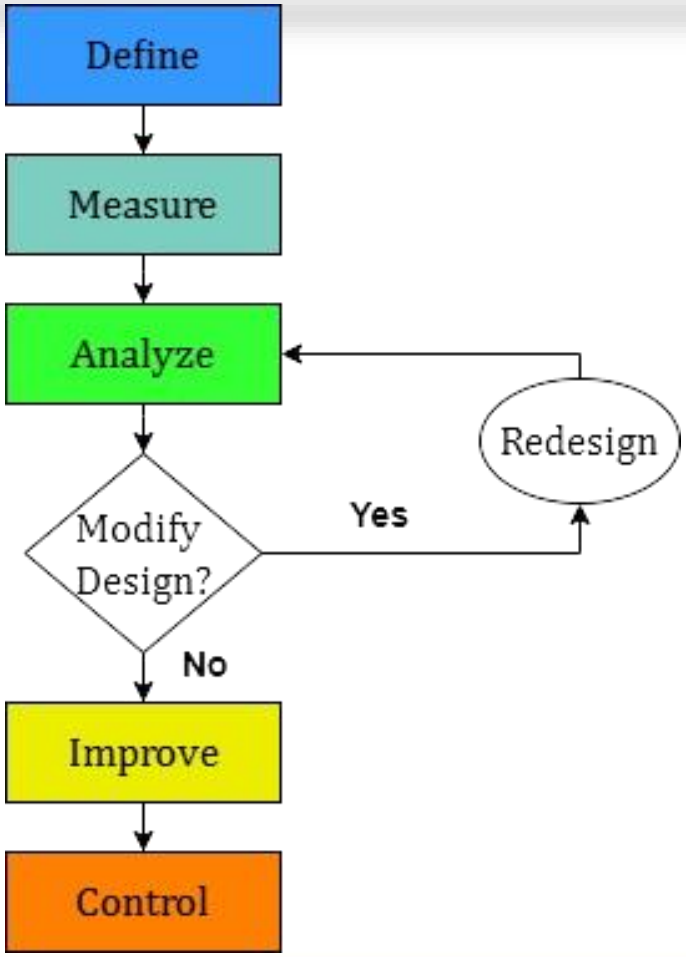
The PDCA (Plan Do Check Act) cycle is an **iterative process** because it involves a series of steps that are continually repeated and refined to achieve continuous improvement. The key to the iterative nature of the PDCA cycle is that, having completed the Act phase, one returns to the Plan phase, building on the knowledge and insights gained in the previous cycle.



Define Measure Analyze Improve Control (DMAIC)

DMAIC is the problem-solving approach that drives Lean Six Sigma. It's a five-phase method—Define, Measure, Analyze, Improve and Control—for improving existing process problems with unknown causes.





Define Measure Analyze Improve Control (DMAIC)



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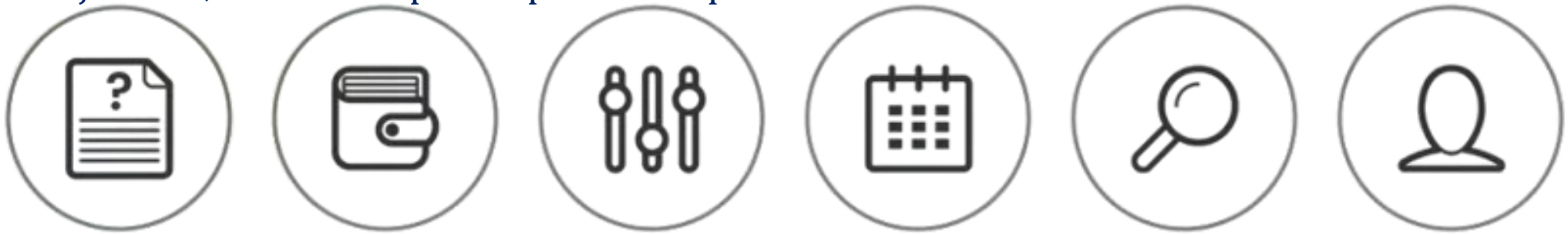
Define – DMAIC Phase 1

The Define phase is the first step in the Lean Six Sigma improvement process.

- In this phase, the project team creates a Project Charter, maps the process at a high level, and identifies the needs of customers related to the process.
- The team also conducts Process Walks (direct observations) and engages in discussions with individuals involved in the process to gain a deeper understanding of how it operates.
- Before proceeding to the Measure phase, the team ensures that the project focus is clearly defined and aligned with the organization's goals.

Define – DMAIC Phase 1

A **Project Charter** is a dynamic document for an improvement team that defines the existing problem, the objectives, and the scope of a process improvement initiative.



PROBLEM STATEMENT
The problem captured in the form of a measurement

BUSINESS CASE
The business reasons for doing the project

GOAL STATEMENT
The target of the process measurement

TIMELINE
When each project phase will be completed

SCOPE
What's in and what's out of the project

TEAM MEMBERS
The people who will participate in the project

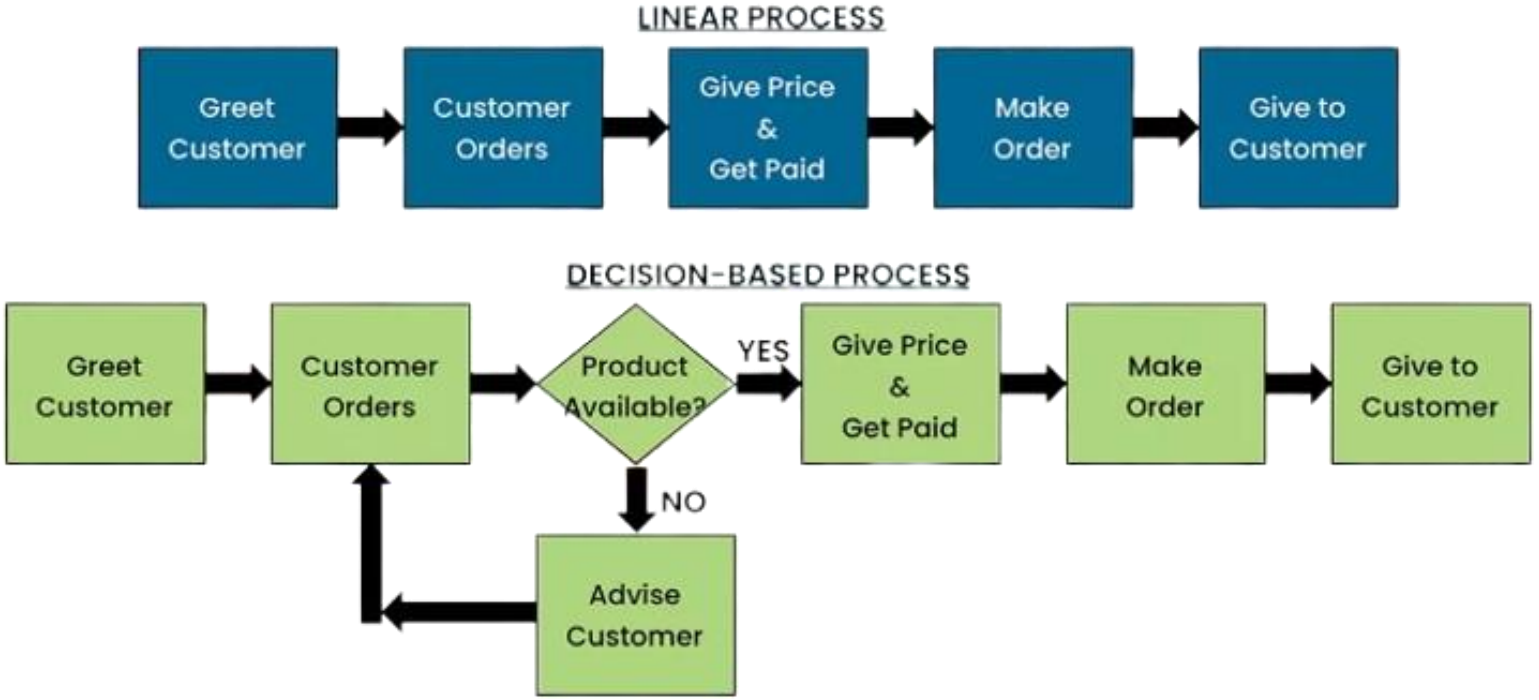
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Measure – DMAIC Phase 2

The **Measure** phase is crucial in a project as it provides key indicators of process health and helps identify where issues are occurring. As the team collects data, they focus on **process completion time** or the **quality of the output received by customers**.

Before proceeding to the **Analyze** phase, the team defines **measurement metrics** and assesses the **current performance** as a baseline for improvement.

A **Process Map** is a sequential diagram that illustrates the activities required to complete a process.



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Analyze – DMAIC Phase 3

The main focus of this phase is to ensure that hypotheses are tested and validated before implementing solutions. Once the analysis is complete and the root cause has been identified, the team can then proceed to the Improve phase.

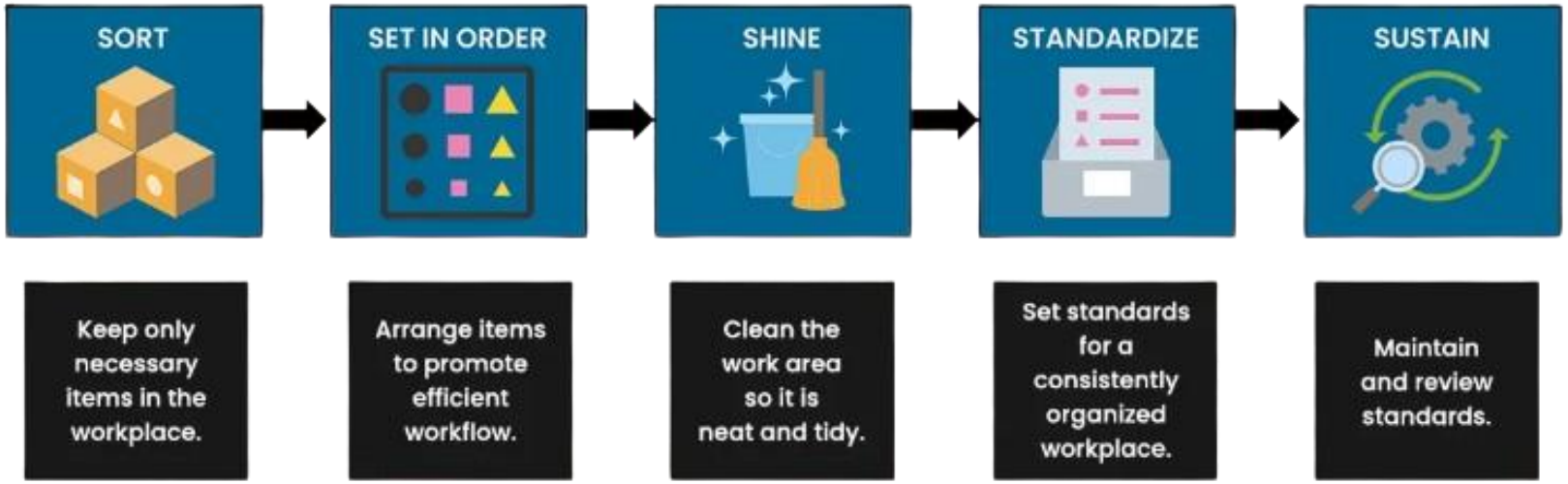
The 5 Whys is a simple analysis technique that goes beyond symptoms by repeatedly asking "why" until identifying the true root cause of an issue.



Improve – DMAIC Phase 4

The **Improve Phase** is where the team refines their countermeasure ideas, pilots process changes, implements solutions, and lastly, collects data to confirm there is measurable improvement. A structured improvement effort can lead to innovative and elegant changes that improve the baseline measure and, ultimately, the customer experience.

5S is a five-step organizing technique designed to create and maintain an intuitive workspace.

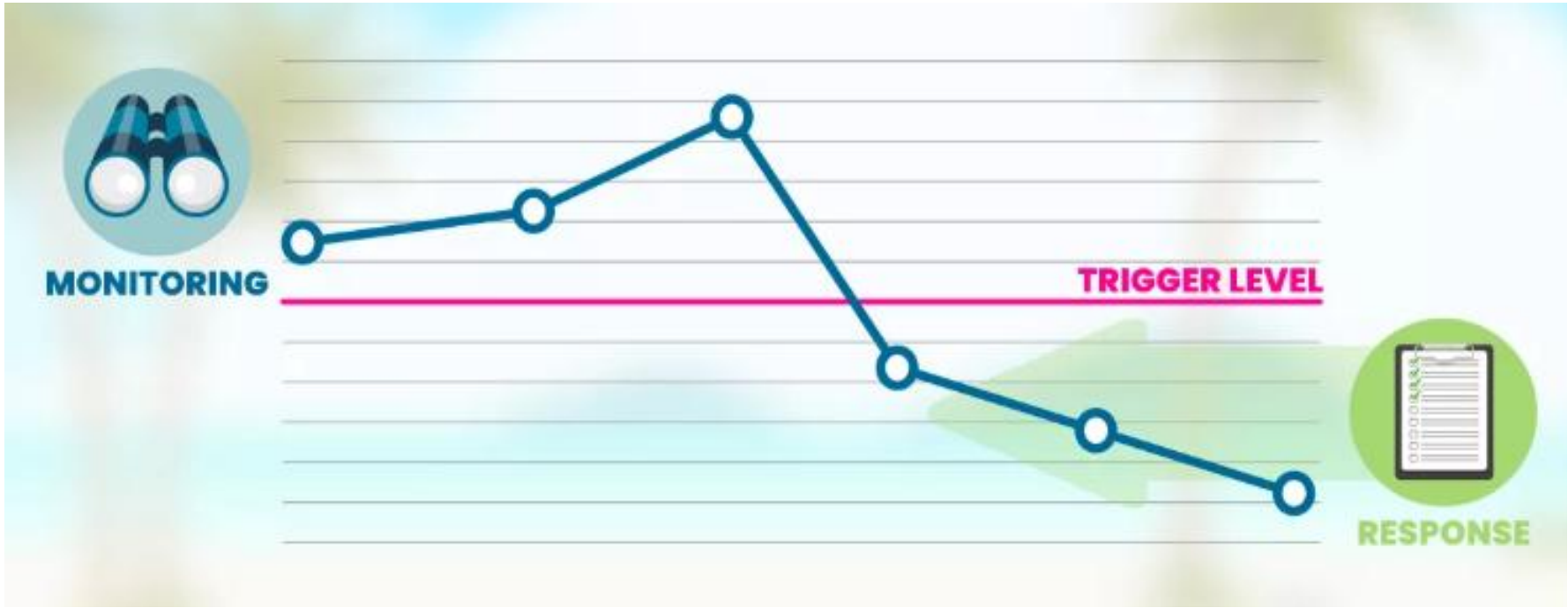


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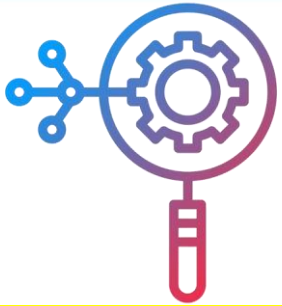
Control – DMAIC Phase 5

In the **Control Phase**, the team develops a Monitoring Plan to track the success of the updated process and crafts a Response Plan in case there is a dip in performance. Once in place, the Process Owner monitors and continually updates the current best method.

The Monitoring & Response Plan evaluates whether a process is functioning as expected and outlines actions to take if it is not.



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Root Cause Analysis

A **root cause** is defined as a factor that caused a nonconformance and should be permanently eliminated through process improvement. The root cause is the core issue—the highest-level cause—that sets in motion the entire cause-and-effect reaction that ultimately leads to the problem(s).
To help the team find answers to why a specific problem arises in your process, RCA (Root Cause Analysis) can be conducted in the following five steps:

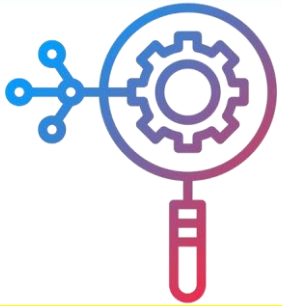
Step 1 – Define the Problem

- What problem is currently occurring?
- Describe the specific symptoms that indicate the existence of this problem!

Step 2 – Collect Data

- Do you have evidence proving that the problem actually exists?
- How long has this problem been present?
- What impact has been felt due to the problem?

At this stage, an in-depth analysis must be conducted before identifying the factors contributing to the problem. To ensure the Root Cause Analysis (RCA) is effective, gather representatives from all relevant departments (from experts to frontline staff) who understand the situation.

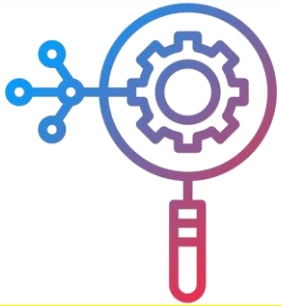


Root Cause Analysis

Step 3 – Identify Possible Causes

- Outline the sequence of events that led to the problem.
- Under what conditions does this problem occur?
- Are there any other issues that arise alongside or as a consequence of the main problem?

At this stage, identify as many potential causes as possible. In many cases, people tend to stop after identifying only one or two causal factors. However, one or two causes are often insufficient to determine the real root cause. RCA is not just about addressing surface-level problems but digging deeper to eliminate the root cause of the entire issue.



Root Cause Analysis

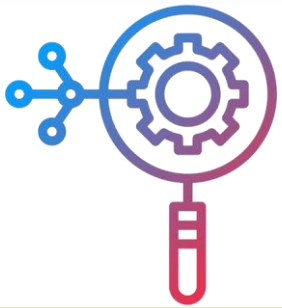
Use the following tools to help identify causal factors:

1. 5 Whys Analysis – Keep asking "Why?" until you reach the fundamental answer.
2. Drill Down – Break down the problem into smaller, more detailed parts to understand the bigger picture.
3. Appreciation – List the available facts and ask, "So what if this happens or doesn't happen?" to identify the most likely consequences of those facts.
4. Cause and Effect Diagram (Fishbone Diagram) – A visual tool that maps out all possible contributing factors to determine where the problem originated.

Step 4 – Identify the Root Cause(s)

- Why do these causal factors exist?
- What fundamental reason truly underlies the emergence of the problem?

Use the same tools from Step 3 to explore the root causes of each identified factor. These tools are designed to encourage deeper exploration at every level of cause and effect.



Root Cause Analysis

Step 4 – Identify the Root Cause(s)

- Why do these causal factors exist?
- What fundamental reason truly underlies the emergence of the problem?

Use the same tools from Step 3 to explore the root causes of each identified factor. These tools are designed to encourage deeper exploration at every level of cause and effect.

Step 5 – Propose and Implement Solutions

- What can be done to prevent the problem from recurring?
- How can the formulated solutions be implemented?
- Who will be responsible for implementing the solution?
- Are there any risks associated with implementing the solution?

Analyze your cause-effect identification process and assess whether changes are needed in other systems. It is also crucial to predict the potential effects of implementing the solution. By doing so, you can prevent or eliminate issues before they surface.



5 Whys Technique

The **5 Whys Problem Solving technique** is a simple process to follow to solve any problem by repeatedly asking the question "Why" (five times is a good rule of thumb), to peel away the layers of symptoms that can lead to the root cause of a problem. This strategy relates to the principle of systematic problem solving.

The 5 Whys Technique and Root Cause Analysis (RCA) are closely related because 5 Whys is one of the methods used in RCA to identify the root cause of a problem. RCA is a systematic approach to determining the fundamental cause of an issue, and 5 Whys serves as one of the techniques that support this process.



5 Whys Technique

Proper uses of the technique:

- To help identify the root cause of a problem
  - A root cause is the most basic reason, which if eliminated, would prevent reoccurrence.
- Provides a framework for a team to work through a more complex problem
- To solve problems as they occur

Improper uses of the technique:

- To emphasize the person or blame: turning the 5 Whys into the 5 Whos
- Making it a tedious, desk intensive project



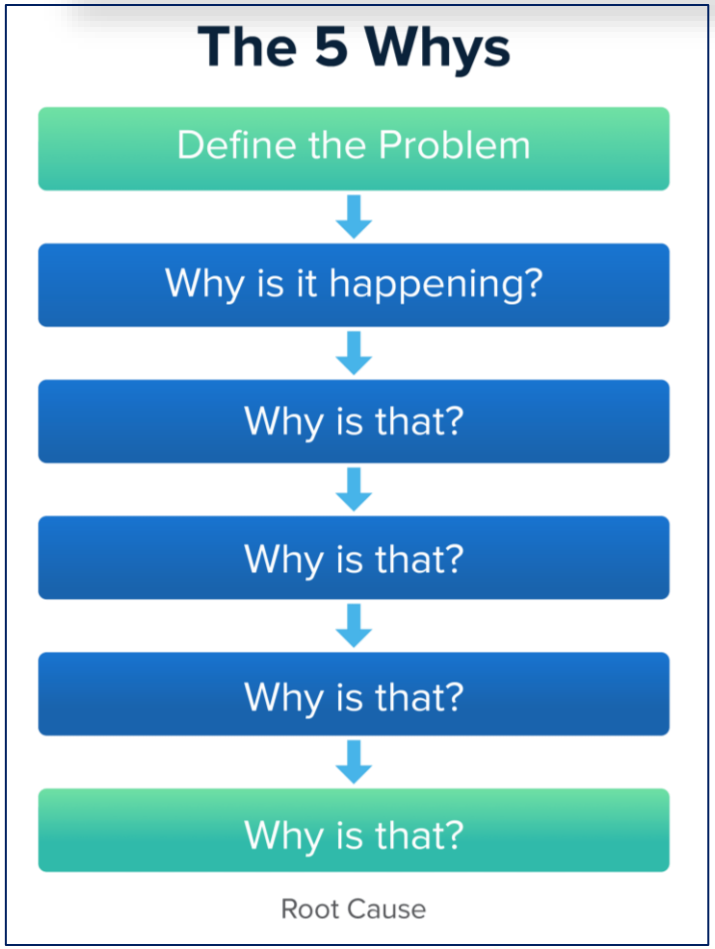
5 Whys Technique

Activity Procedure:

1. Write down the specific problem.
   - Writing the issue helps you formalize the problem and describe it completely.
   - It helps a team focus on the same problem.
   - Always describe the current condition.
   - Use data where possible.
     - Example: Overall customer complaints are up 50%.
2. Ask WHY the problem happens and write the answer down below the problem.
3. If the answer provided doesn't identify the root cause of the problem that you wrote in step 1, ask WHY again and write that answer down.
4. Loop back to step 3 until the team is in agreement that the problem's root cause is identified.



5 Whys Technique



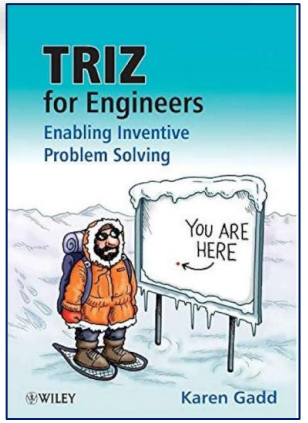
Case Example:

Problem Statement: Your company was unable to get the customer's product request to them on time.

1. Why were you unable to produce the product on time?
   - Because the equipment failed.
2. Why did the equipment fail?
   - Because the circuit board burned out.
3. Why did the circuit board burn out?
   - Because it overheated.
4. Why did it overheat?
   - Because the air filter wasn't changed.
5. Why wasn't the filter changed?
   - Because there was no afternoon preventative maintenance shift scheduled to change it.

Theory of Inventive Problem Solving (TRIZ)

Definition

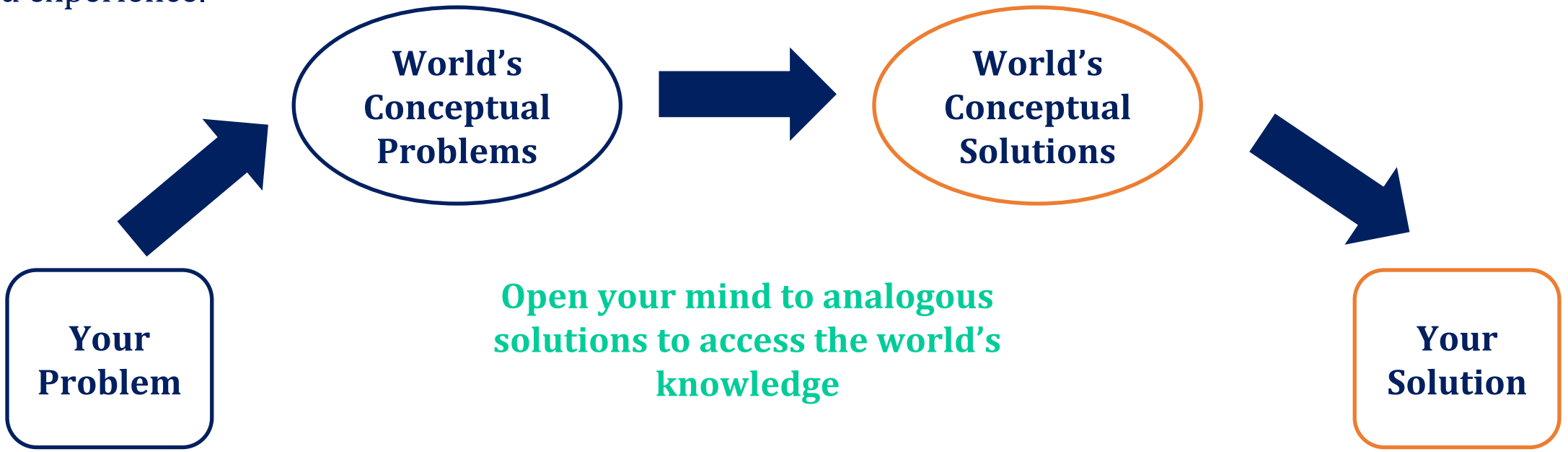


TRIZ is an engineering problem solving toolkit which successfully summarizes past solutions and successes to show us how to systematically solve future problems. TRIZ comes from Russia, initially and primarily the work of Genrich Altshuller, a great engineer and inventor, perhaps one of the greatest engineers of the twentieth century, whose work helps all other engineers.

TRIZ focuses our problem understanding to the particular, relevant problem model and then offers conceptual solutions to that model. Good engineers reduce wasted time with TRIZ as they head straight for the valid solutions and use their valuable time to define their problem accurately, find all the solutions to that problem and then develop those solutions.

Prism of TRIZ

The simplest TRIZ flowchart is the Prism of TRIZ which is a simple convergent /divergent tool for finding existing solutions, converging our real - life problems to a conceptual problem with conceptual answers and diverging the conceptual solutions back to real - life answers by adding back the relevant knowledge and experience.



Theory of Inventive Problem Solving (TRIZ)

Where Do We Start with TRIZ? Which Tools When?

There are simple stages which we can apply to any problem:

- ❑ Defining the Ideal for setting the direction of the problem solving and the scope for stakeholders.
- ❑ Problem Understanding – mapping the problem context in Time and Scale and capturing the relevant facts for both requirements and the system.
- ❑ Finding, creating and combining all the previous, spontaneous and known Solutions and Concepts. Filling up a ‘ Bad Solution ’ Park using everyone involved in the problem.
- ❑ Choosing/developing/transforming bad solutions into better solutions using TRIZ. Map all TRIZ transformed solutions in 9 boxes.

Thank you



UNIVERSITAS
GADJAH MADA

Solution Strategies

Utilization, Nullification, Elimination

Problem Solving Methods and Modeling

Dr. Indriana Hidayah, S.T., M.T.

Unified Structured Inventive Thinking (USIT)

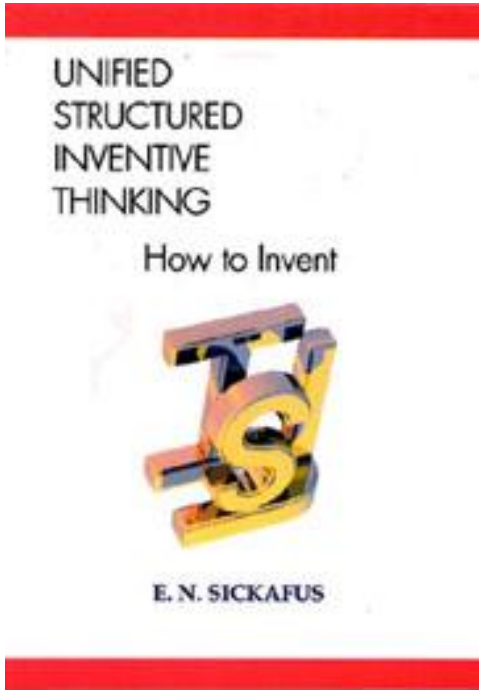


What is USIT

Unified Structured Inventive Thinking is a problem-solving method developed as a simplification of TRIZ.

Solution Strategies

In USIT, Sickafus also developed a problem-solving strategy based on **Utilization, Nullification, and Elimination**.



Utilization

Utilizing existing resources within the system to solve a problem without adding new elements.

Nullification

Neutralizing unwanted effects by reversing or eliminating their impact.

Elimination

Removing parts of the system that cause the problem or finding a solution that eliminates the need for certain elements

Utilization

The approach of utilizing existing resources within a system to solve problems without adding new elements. Aligns with the Closed-World Method principle, where solutions must be found using the minimal set of objects within the system.

In Closed-World Analysis, existing objects are analyzed to identify attributes and functions that can be utilized to improve the system without adding new components



Sample Case

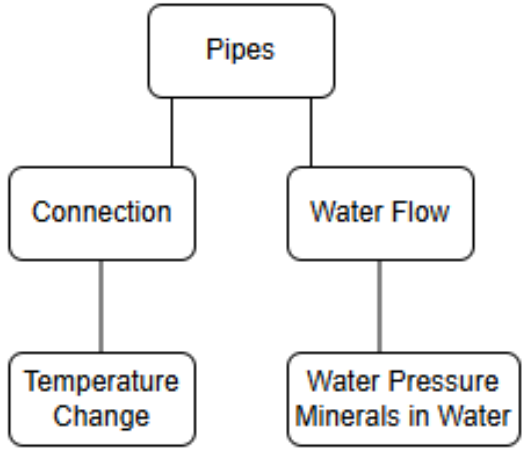
Reducing Water Leaks in Pipes Without Adding New Materials

With the Closed-World Method and Utilization approach, we seek solutions without introducing new materials from outside the system.

Case Resolution

1. **Minimal Set of Objects**
   - Pipe
   - Joint
   - Water Flow
   - Pressure in Pipe
2. **Utilization Approach**
   - Utilizing the water pressure itself
   - Taking advantage of the pipe's temperature changes
   - Harnessing the minerals in the water

3. **Closed-World Diagram**



Using water pressure to seal leaks

By adjusting the pressure within the pipe, small leaks can be sealed by the back pressure of the water itself.

Utilizing pipe temperature changes

If the pipe's temperature increases, the material around the joints will expand, naturally sealing the leak.

Harnessing minerals in the water

If the water contains lime or other minerals, these minerals can accumulate at the leak point and naturally seal it over time.

Exercise Utilization



Railway track bearings experience wear due to high friction between train wheels and the rails. Typically, to reduce friction and wear, external lubricants such as grease or oil are used



Find a solution without adding external lubricants, but by utilizing the existing materials within the system.

Nullification

Neutralizing the negative effects of a problem without eliminating its source. This closely resembles the concept of Transduction, where changes are made by redirecting problematic attributes or functions to an alternative pathway.

Case Example:

Overcoming Caffeine Side Effects Without Stopping Coffee Consumption



With the **Transduction technique** and **Nullification solution**, coffee consumption will not be stopped, but its negative effects will be neutralized, allowing the benefits of caffeine to be enjoyed without disruption.

Case Resolution

1. **Problem:** Caffeine is still consumed
2. **Negative effects:** Side effects such as heart palpitations and increased stomach acid.
3. **Nullification Approach**

   Not stopping coffee consumption, but neutralizing its negative effects to continue enjoying the benefits of caffeine without disruption.



1. Consume magnesium-rich sources (e.g., bananas, avocados, or magnesium supplements) to neutralize excessive stimulation on the heart.
2. Drink coffee after meals to reduce its impact on the stomach or Drink a glass of water after coffee to replenish lost fluids.

Exercise Nullification



1. **When an aircraft lands, its wheels experience a significant impact force, causing strong shocks that can:**
   - Endanger passengers due to severe jolts.
   - Cause premature wear on the wheels and aircraft frame.
   - Increase maintenance costs due to high stress on the suspension system



Use the Nullification approach to Neutralizing or reducing the negative effects of a problem without eliminating its source.

Elimination

Eliminating the source of the problem by removing elements that cause undesirable effects. This aligns with the concepts of Pluralization and the Closed-World Method, which emphasize simplifying the system by eliminating unnecessary components.

Sample Case

Eliminating Physical Textbooks with E-Books and Learning Management Systems (LMS)



The conventional solution is usually to reprint books in a lighter or more affordable edition, but this still does not address the root of the problem

Case Resolution



1. **Main Cause:** The use of physical textbooks, which are inflexible and expensive.
2. **Negative Impact:** Increased student burden, high costs, and limited access.
3. **Elimination Approach:** Instead of making textbooks lighter or cheaper, we completely remove physical books and replace them with e-books and a Learning Management System (LMS).

1. Eliminating the need for physical textbooks by replacing them with e-books (Easy Access).
2. Utilizing a Learning Management System (LMS) to store and manage learning materials online.
3. Enabling content updates without the need to reprint books
4. Easily accessible from any device (laptop, tablet, or smartphone).

Exercise Elimination



A company manages its own physical servers to run applications and store data. However, they face several issues:

- High operational costs
- Lack of flexibility
- High downtime risk



Use the **Elimination** approach to eliminate the negative impact of shocks without adding new materials or components.

CONCLUSION

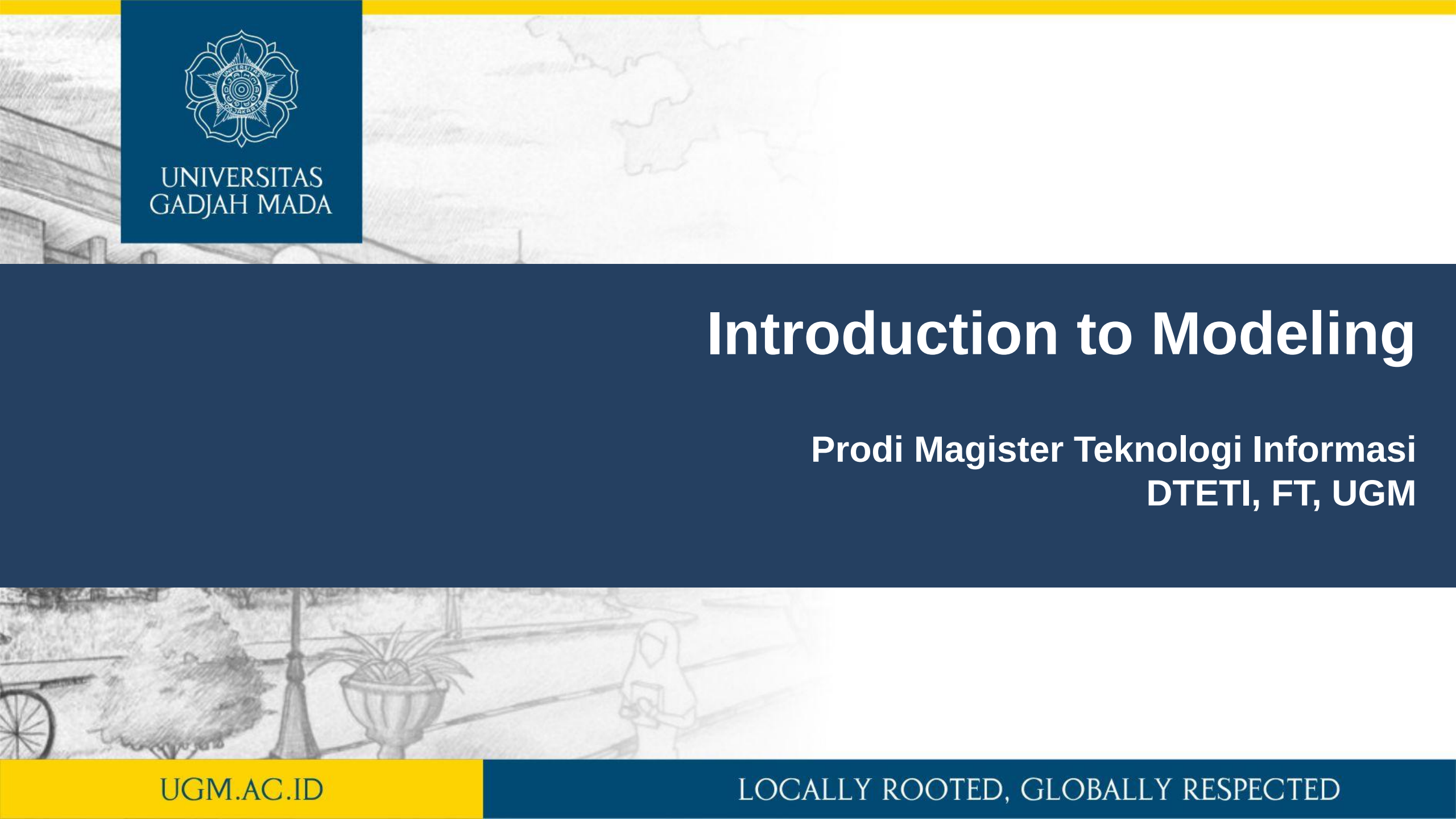
These three strategies offer a more efficient approach to problem-solving compared to conventional solutions. The choice of strategy depends on the nature of the problem:

If existing resources can be utilized → **Use Utilization**

If negative effects can be eliminated without changing the cause → **Use Nullification**

If the root cause can be completely removed → **Use Elimination**

Any Questions?



Introduction to Modeling

Prodi Magister Teknologi Informasi
DTETI, FT, UGM

Are problem-solving and modeling related?

What is problem-solving?

What is modeling?

What is modeling?

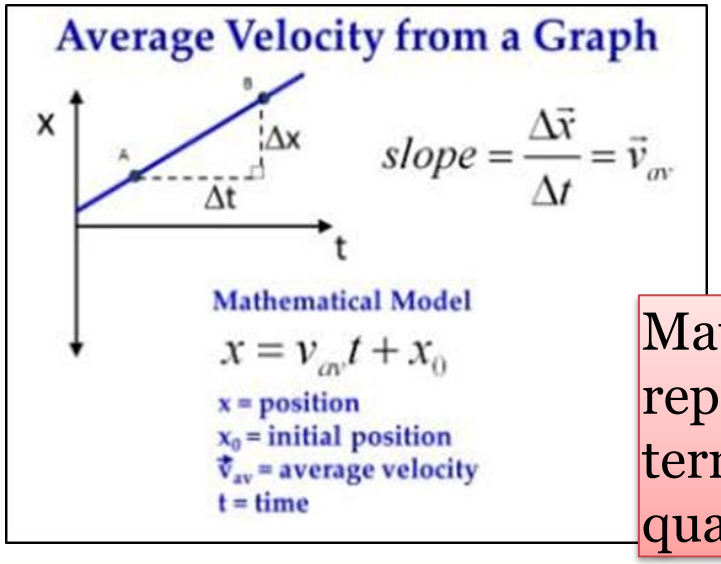
Modeling is the process of making/creating a model.

A model is

- A simpler representation of something (item, phenomenon, system, process).
- Models can also be said to be replicas; they can be physical or mathematical.







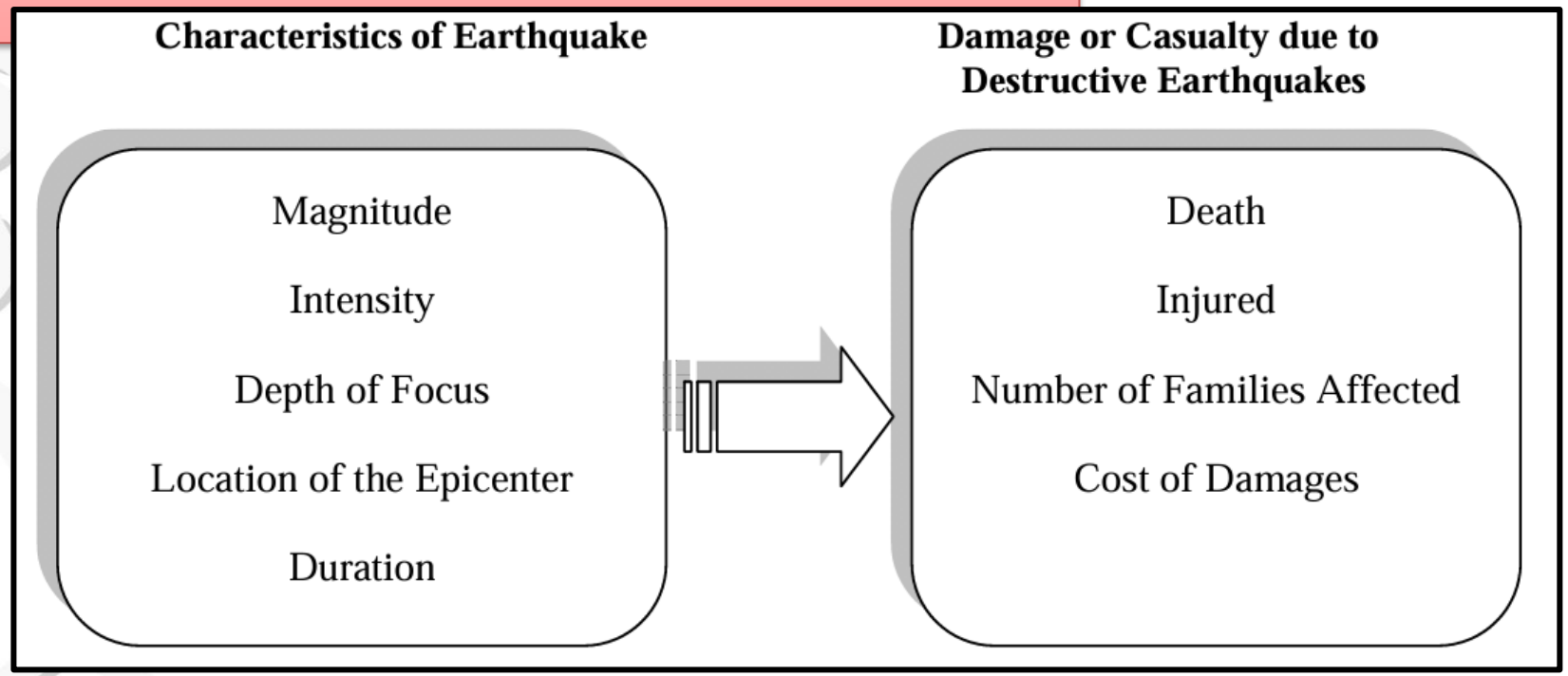
Mathematical modeling is representing a system in terms of logical and quantitative relationships.

Are problem-solving and modeling related?

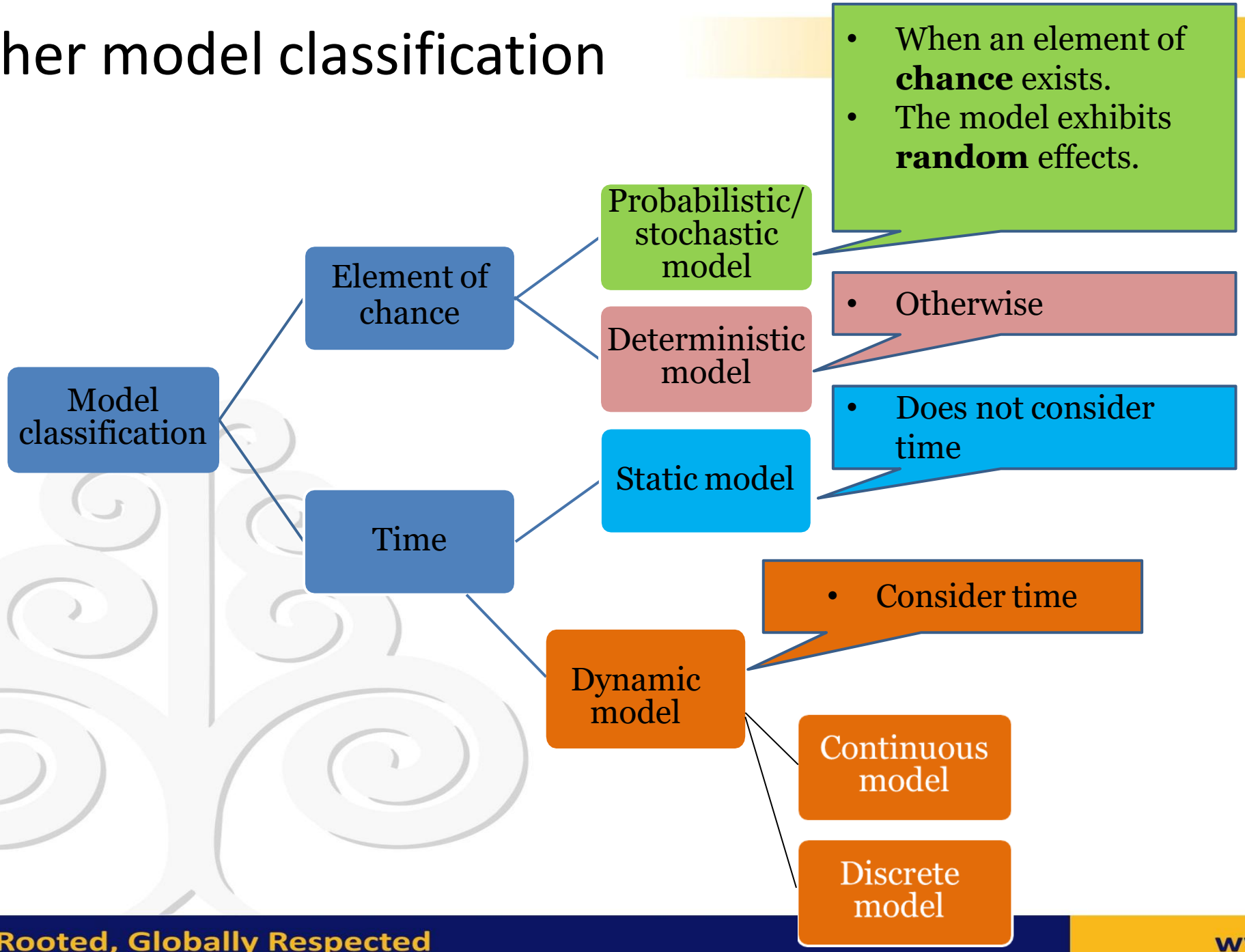
- We use modeling to understand the problem by creating simplified version of the real-life systems.

- We can use modeling to **analyze** complex, real-world problems to **predict** what might happen with some course of action.
- Example:
  - Use math models to simulate effects of earthquake (making quantitative predictions of infrastructural damages, response, and recovery to help minimize damage, death, and injury).



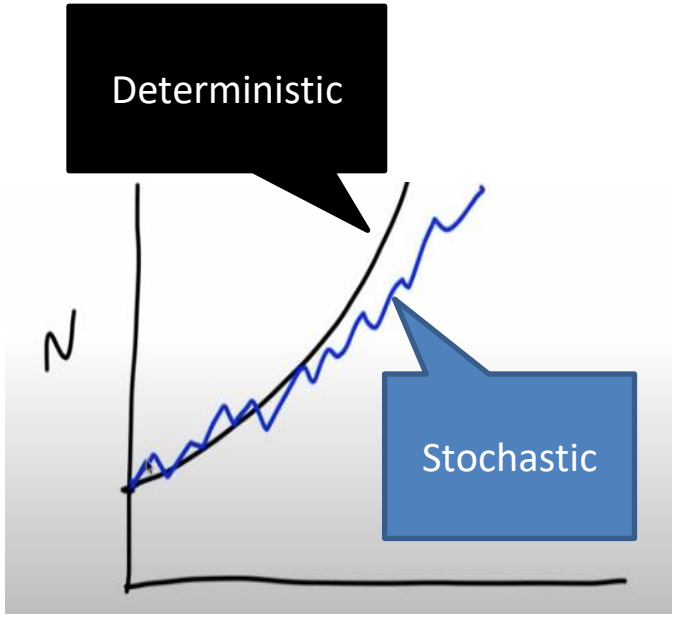


Another model classification



Examples of stochastic and deterministic models

- Stochastic models have elements of randomness.
  - The same parameter values and the same initial conditions can lead to different outcomes.
- Let's compare deterministic with stochastic:
  - If population growth could be modeled by
    - $N(t+1) = N(t)r$, where r is the growth rate → deterministic model
    - $N(t+1) = N(t)r(t)$, where r(t) indicates that the growth rate may change over time for some reason (random) → stochastic model



Static model represents of a system at a particular time, or one that may be used to represent a system in which time simply plays no role.

- Static (steady-state) models consider systems in equilibrium, thus are time-invariant.
- Static models are more structural than behavioral.
- Static modeling is more rigid than dynamic modeling because it is a time-independent view of the system.
- Example:
  - replica of sea or building or car
  - calculation of mechanical stress in a bridge

Contoh model dinamis

Dynamic model represents a system as it evolves over time, such as a conveyor system in a factory.

- Dynamic modeling is flexible because it can change over time because it shows what an object does with many possibilities that may arise in time.
- Example:
  - Math model of mechanical stress in a bridge that consider effects of **moving** traffic, wind, earthquakes, or other things that vary in time.

Continuous and discrete model

Definitions In a **continuous model**, time changes continuously, while in a **discrete model**, time changes in incremental steps.

- We often develop discrete computer models of continuous situations.
- Example?

Steps of Modeling Process - Cyclic

Analysis of the problem

Formulate model

- Gather data
- Make simplifying assumptions and document them
- Determine variables and unit
- Establish relationship between variables
- Determine equation and functions

Steps of Modeling Process - Cyclic

Solve the model

Verify and interpret the model solution

Steps of Modeling Process - Cyclic

Report on the model

- Analysis of the problem
- Model design
- Model solution
- Result and discussion

Maintain the model

Exercise

1. Compare and contrast the modeling process with the scientific method: Make observations; formulate a hypothesis; develop a testing method for the hypothesis; collect data for the test; using the data, test the hypothesis; accept or reject the hypothesis.
2. Compare and contrast the modeling process with the software life cycle: analysis, design, implementation, testing, documentation, maintenance.

What is simulation?

Imitation of the operation of a real-world process or system over time.

→ need to develop a simulation **model.**

- Examples: …

- The process of
  - designing a mathematical or logical model of a real-system and then
  - conducting computer-based experiments with the model to
    - describe, explain, and predict the behavior of the real system.
- The actual running of the model system to gain insight into its performance.
- The process which simulates in the laboratory or on the computer, the actual scenario as close to the system as possible.

Simulation benefit: predict the system behavior & suggest improvements in it

Use simulations if

- Not feasible to do actual experiments
- System does not exist (VM ware)
- Cost of actual experiments prohibitive (discourage)
  - Money
  - Time
  - Danger
- Want to test alternatives

Actual Trial vs Simulation

e.g. damage from weapon

- **Actual trial** (some says actual simulation)

  => drop the weapon on the target and see its end effects.

- **Computer/laboratory simulation**

  => conducting trials on computer/laboratory, which simulates the end effects of the weapon on the target.

Any questions?

Chapter 2

SYSTEM DYNAMICS PROBLEMS WITH **RATE PROPORTIONAL TO AMOUNT**

Metode Penyelesaian Masalah dan Pemodelan

DTETI, FT, UGM



SYSTEM DYNAMICS PROBLEMS WITH RATE PROPORTIONAL TO AMOUNT

Growth and decay (Module 2.2, Module 2.3)

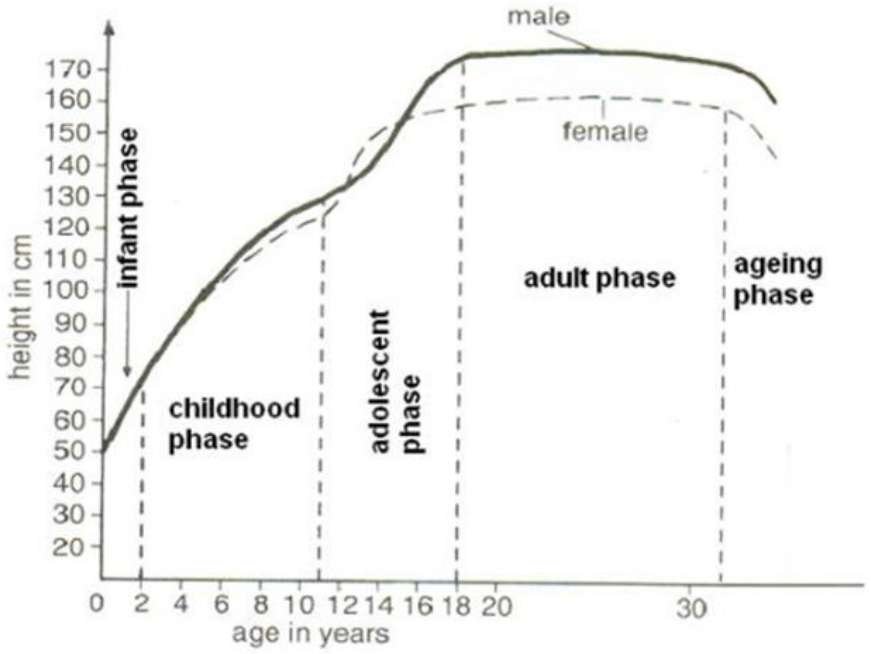
Drug dosage (Module 2.5)

Rate of change is proportional to the amount present.

$$\text{average velocity} = \frac{\Delta s}{\Delta t} = \frac{186.7\ \text{km} - 16.1\ \text{km}}{2\ \text{h} - 0\ \text{h}} = \frac{170.6\ \text{km}}{2\ \text{h}} = 85.3\ \text{km/h}$$

Example

Rate of change: example



Human growth curve

- Growth rate = change of height / change of time
- Growth rate in each phase?



SYSTEM DYNAMICS PROBLEMS WITH **RATE PROPORTIONAL TO AMOUNT**

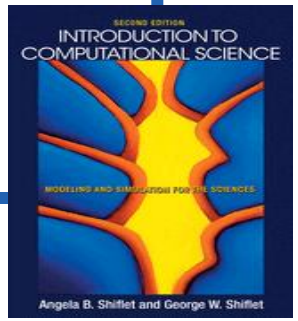
Modeling Growth and Decay

Why modelling growth and decay?

- There are many problems in everyday life that require growth and decay models for prediction, planning and improvement purposes..
  - Population (people, animals, bacteria)
  - Viewers of viral video
  - Radioactivity
  - Toxic waste left in a lake
- There are many other examples of problems, everything that basically concerns the rate of change, whether positive or negative..

- Example:
  - Rate of change of population $\frac{dP}{dt} \propto P$

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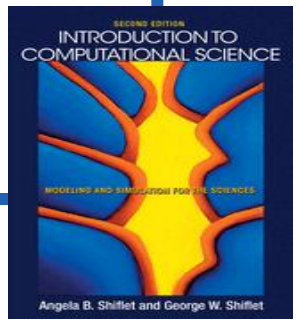
Module 2.2
Unconstrained Growth and Decay

Angela B. Shiflet and George W. Shiflet

Wofford College

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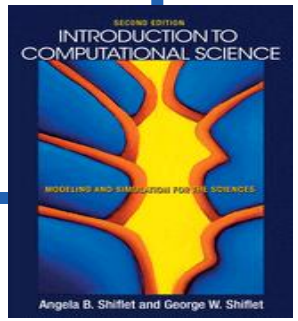
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- Assume we have a population in which
  - no one comes or goes;
  - births and deaths are the only causes of population change.
- There are no constraints such as competition for food or predation.
  - This means there are no restraining factors

    → ***unconstrained*** *growth model*

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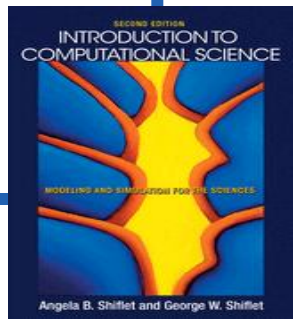
Example of unconstrained growth

- Population growth without constraints
- *Rate of change of population is directly proportional to number of individuals in the population (P)*
- Differential equation $\frac{dP}{dt} = rP$, where ***r*** is growth rate
- $\frac{dP}{dt} = r_1 P - r_2 P = (r_1 - r_2)P$

Malthusian model
(simple exponential growth model)

Model untuk unconstrained growth

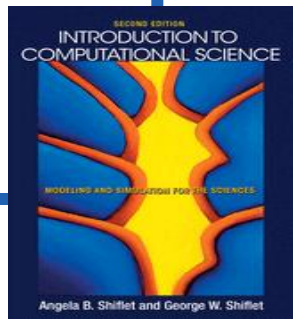
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Example of unconstrained growth

- Differential equation $\frac{dP}{dt} = rP$, where ***r*** is growth rate

Definitions A **differential equation** is an equation that contains one or more derivatives. An **initial condition** is the value of the dependent variable when the independent variable is zero. A **solution** to a differential equation is a function that satisfies the equation and initial condition(s).



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Analytic Solution

- For example, given that growth_rate r=0.1, then

$$\frac{dP}{dt} = 0{,}1P$$

$$\frac{1}{P}dP = 0{,}1dt$$

$$\int \frac{1}{P}dP = \int 0{,}1\,dt$$

$$ln|P| = 0{,}1t + C$$

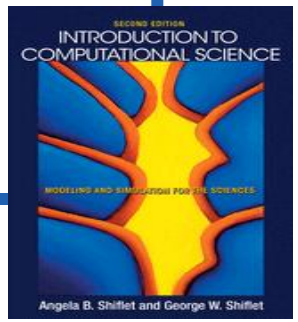
$$e^{\ln|P|} = e^{0.10t+C}$$

$$|P| = e^{0.10t}e^{C} = A\,e^{0.10t}$$

where $A = e^{C}$. Solving for P, we have

$$P = (\pm A)e^{0.10t}$$

$$P = ke^{0.10t}$$



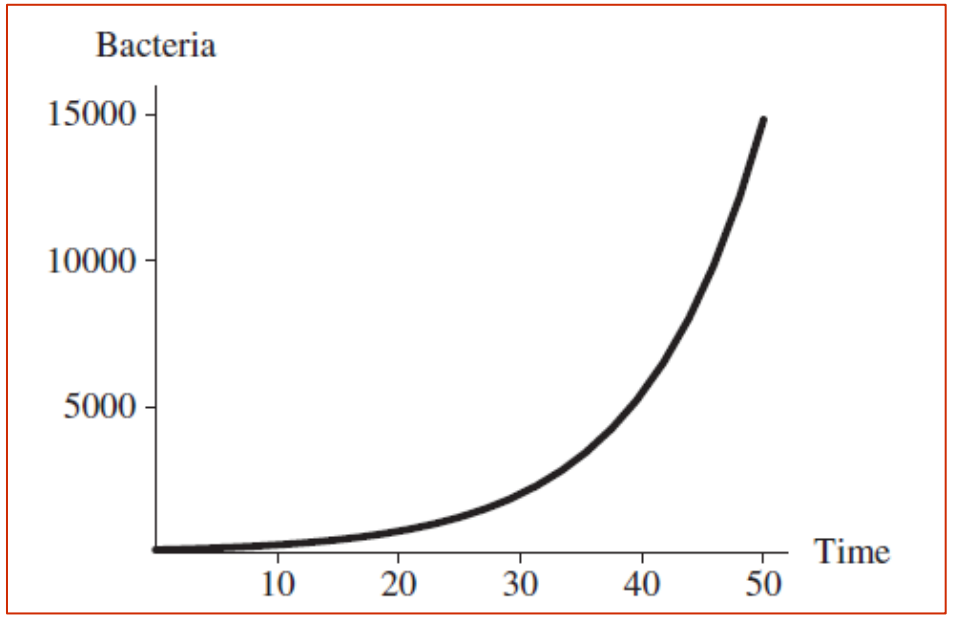
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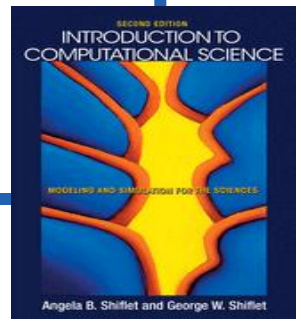
Analytic Solution

- In this example, constan k is the initial population of a particular bacteria. Therefore,

$$P = P_0 e^{rt} = 100\, e^{0.1t}$$

The bacteria population growth is plotted





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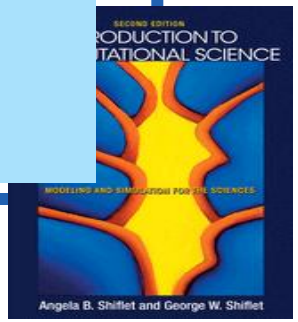
In general, the solution to

$$\frac{dP}{dt} = rP \quad \text{with initial population } P_0$$

is

$$P = P_0 e^{rt}$$

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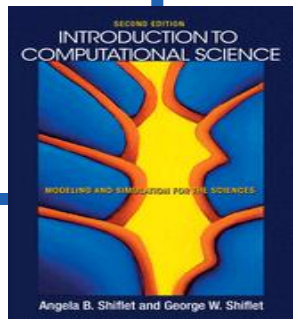
For the same case

- It is known that the initial population of a type of bacteria is 100 and the instantaneous growth rate is 10% = 0.10/hour. So we have a differential equation

  $dP/dt = 0.10Po$ where Po = 100.

  Differential equation

- Can determine with a **computer** algebra system?



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Finite difference equation

Definition A **finite difference equation** is of the following form:

(new value) = (old value) + (change in value)

Such an equation is a discrete approximation to a differential equation.

$$P_t = P_{t-\Delta t} + (P_{t-\Delta t} * r) * \Delta t$$

Finite difference equation

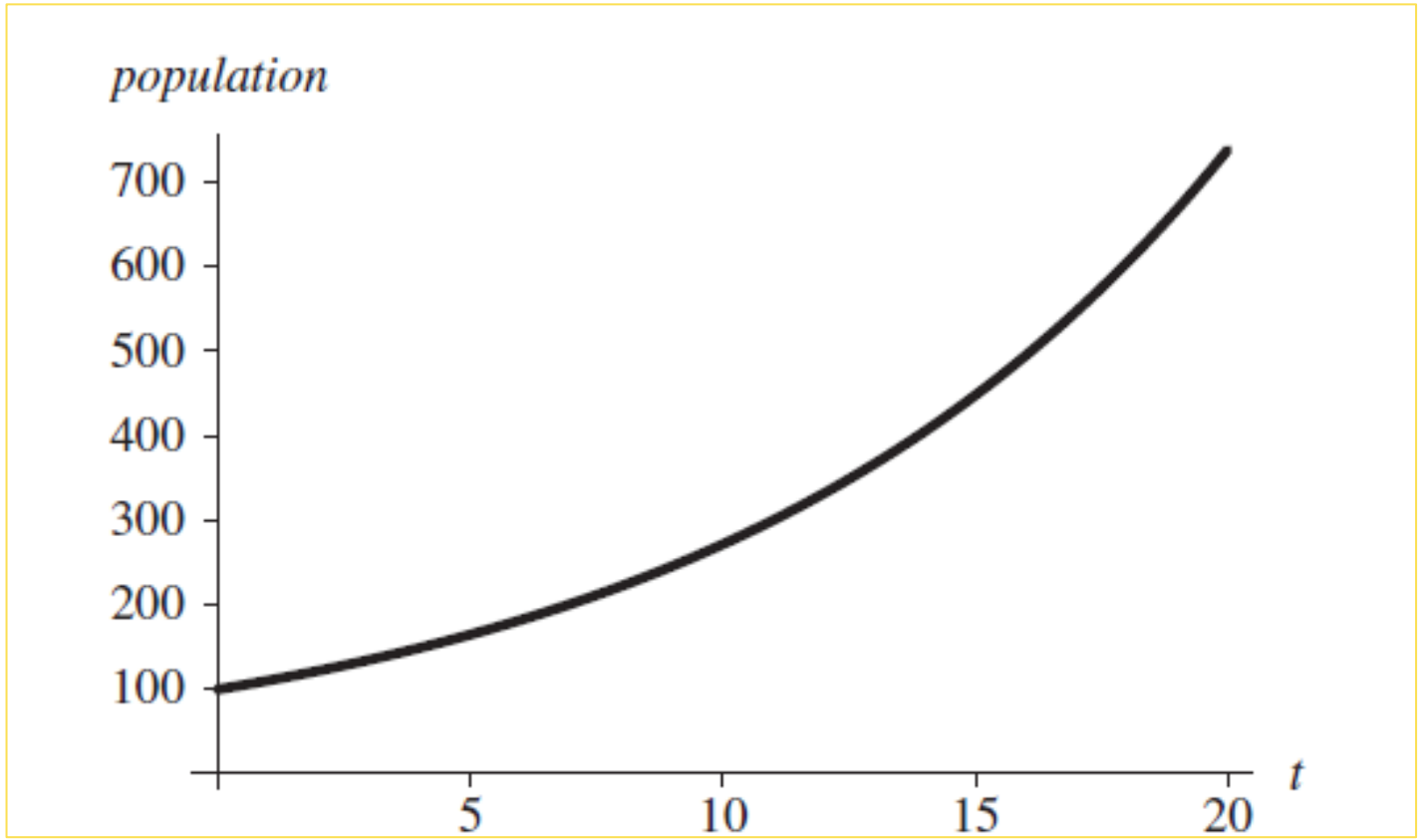
Table 2.2.1
Table of Estimated Populations, Where the Initial Population is 100, the Continuous Growth Rate is 10% per Hour, and the Time Step is 0.005 h

t	$population(t)$	=	$population(t-\Delta t)$	+	$(growth)$	*	Δt
0.000	100.000000						
0.005	100.050000	=	100.000000	+	10.000000	*	0.005
0.010	100.100025	=	100.050000	+	10.005000	*	0.005
0.015	100.150075	=	100.100025	+	10.010003	*	0.005
0.020	100.200150	=	100.150075	+	10.015008	*	0.005
0.025	100.250250	=	100.200150	+	10.020015	*	0.005
0.030	100.300375	=	100.250250	+	10.025025	*	0.005
0.035	100.350525	=	100.300375	+	10.030038	*	0.005
0.040	100.400701	=	100.350525	+	10.035053	*	0.005

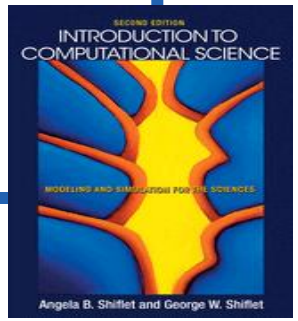
$$P_t = P_{t-\Delta t} + (P_{t-\Delta t} * r) * \Delta t$$

Rule of Thumb Although the simulation takes longer because of more computation, it is usually more accurate to use a small step size (Δt), say, 0.01 or less.

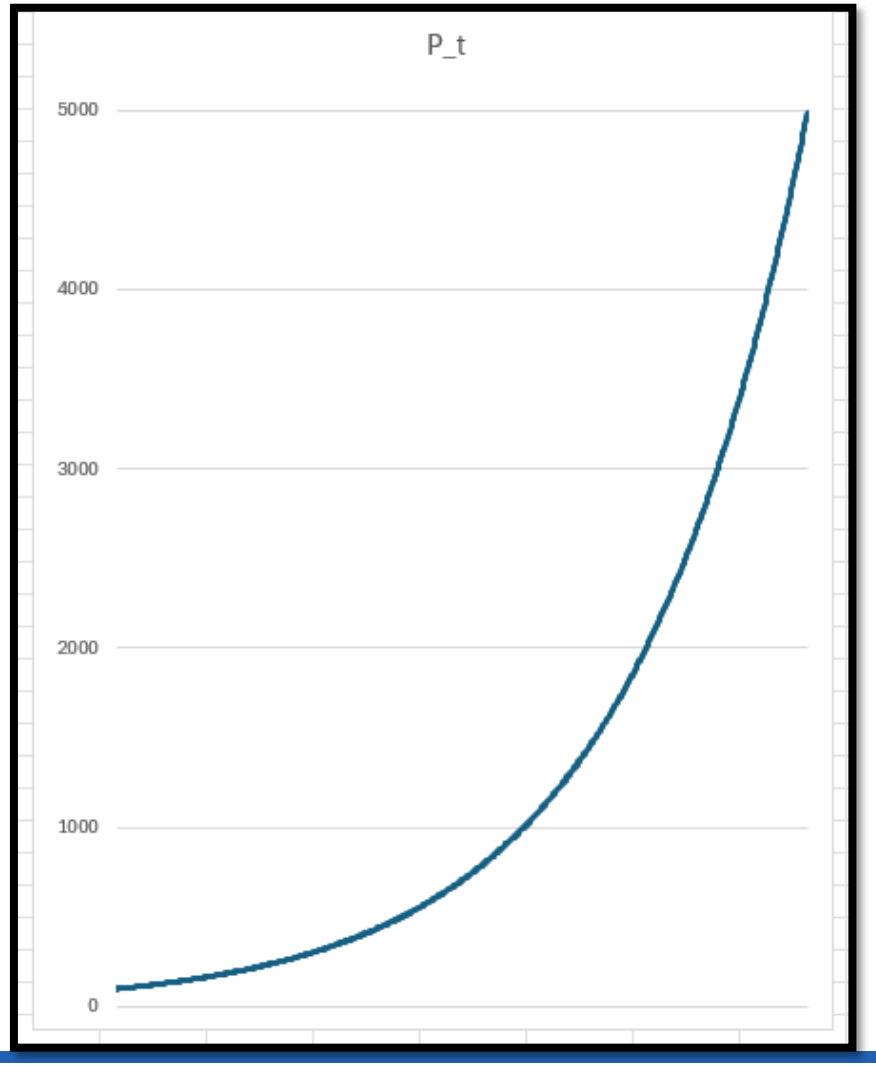
Graph of population versus time (hours)

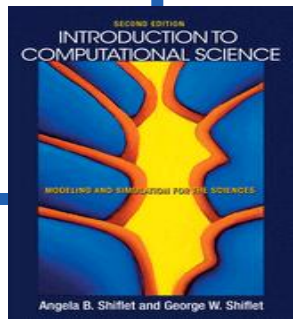


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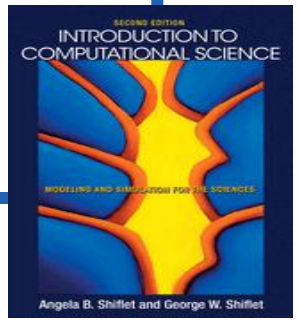


1	Diketahui	r =	0.1		
2					
3	**t**	**P_t**	**P_t-dt**	**G**	**dt**
4	0	100			
5	0.05	100.5	100	10	0.05
6	0.1	101.003	100.5	10.05	0.05
7	0.15	101.508	101.003	10.1003	0.05
8	0.2	102.015	101.508	10.1508	0.05
9	0.25	102.525	102.015	10.2015	0.05
10	0.3	103.038	102.525	10.2525	0.05
11	0.35	103.553	103.038	10.3038	0.05
12	0.4	104.071	103.553	10.3553	0.05
13	0.45	104.591	104.071	10.4071	0.05
14	0.5	105.114	104.591	10.4591	0.05
15	0.55	105.64	105.114	10.5114	0.05
16	0.6	106.168	105.64	10.564	0.05
17	0.65	106.699	106.168	10.6168	0.05
18	0.7	107.232	106.699	10.6699	0.05
19	0.75	107.768	107.232	10.7232	0.05
20	0.8	108.307	107.768	10.7768	0.05
21	0.85	108.849	108.307	10.8307	0.05
22	0.9	109.393	108.849	10.8849	0.05
23	0.95	109.94	109.393	10.9393	0.05





Using modeling and simulation software



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Model representation with several different tools

- Figure a) was created using a modeling tool called Berkeley Madonna ®, b) was created using a modeling tool called STELLA ®, c) was created using Vensim PLE ®, and d) was in the format used in the textbook (Shiflet, 2014).

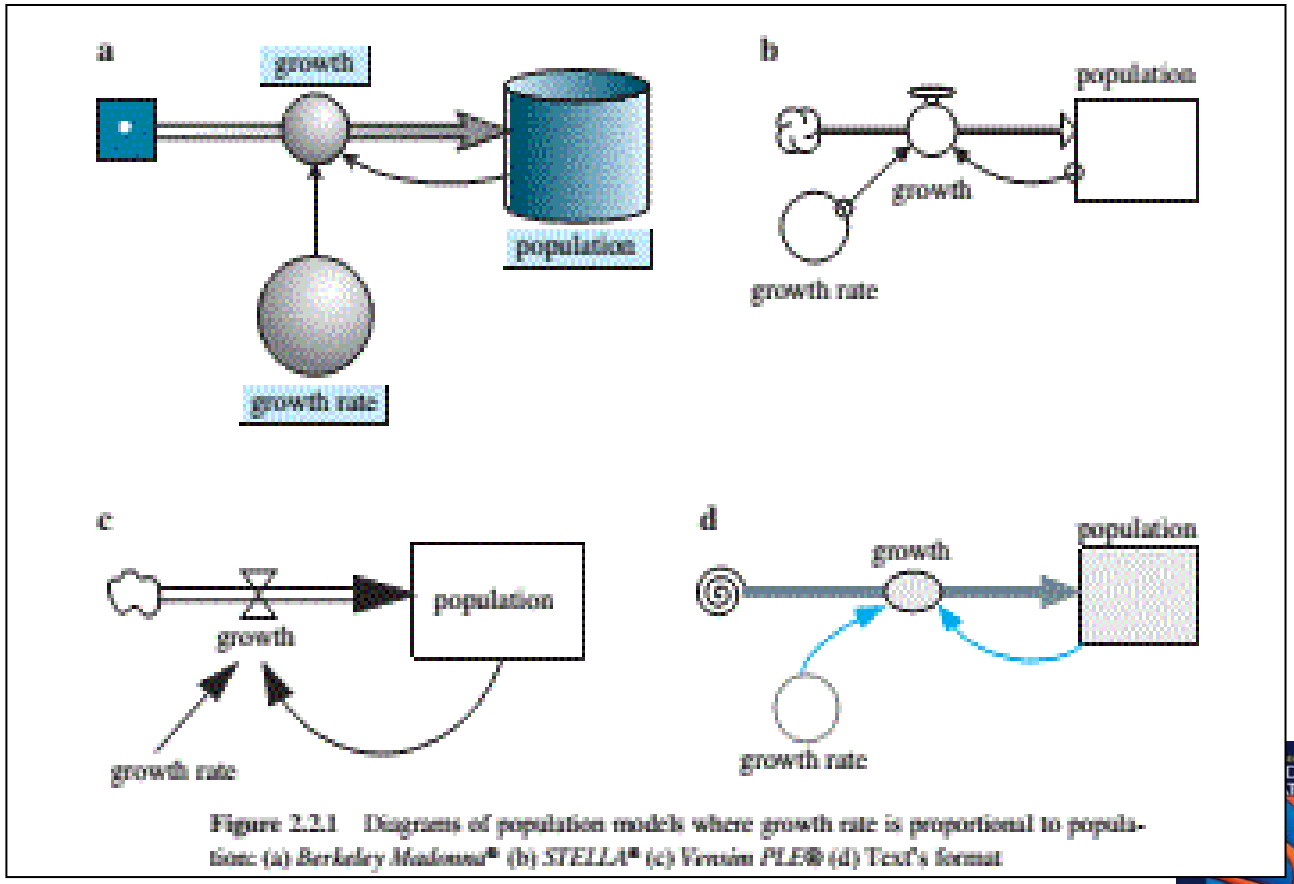
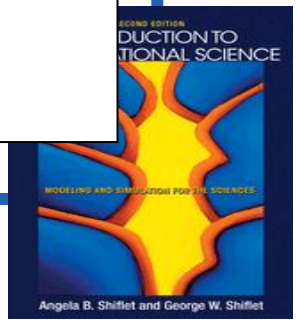


Figure 2.2.1 Diagrams of population models where growth rate is proportional to popula-
tion: (a) *Berkeley Madonna*® (b) *STELLA*® (c) *Vensim PLE*® (d) Text's format



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Examples of modeling and simulation tools: Insight Maker

Home

PRIMITIVE | FLOWS/TRANSITIONS | LINKS | SETTINGS | SAVE | EDIT | SIMULATE

Insight Maker Basics

Click me to Clear this Demo Model

Step 1. Build your Model

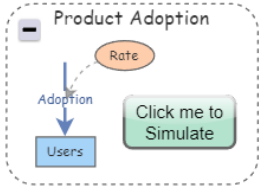
Diagram

Use an extensive set of drawing tools to create causal loop diagrams, flow charts or rich pictures. Build the diagram that communicates your insights and mental models.

This diagram is an Insight Maker model. Feel free to change it. Each picture, textbox or other element is called a "Primitive". You can click and drag primitives to move them or edit their contents in the sidebar.

Simulate

Add equations to your model to simulate future behavior and changes. Use a powerful equation language and large library of built-in functions to simulate System Dynamics or Agent Based Models. View and explore results directly within Insight Maker.



Analyze

Analyze your model results. Study the trends to understand the dynamics of your system. Use Insight Maker's Sensitivity Testing Tool to understand the effect of randomness. Apply Insight Maker's Optimization Tool to maximize or minimize a goal. Utilize Insight Maker's scripting capabilities to take complete control of your analysis.

Step 2. Share your Insights

Insight Maker models are web pages, so sharing them is as easy as emailing a link or posting it to Twitter or Facebook.

If you have your own web page or blog, you can embed Insight Maker models right in your page.

Insight Maker also provides a great way to explain your model: Stories.

Stories create a walkthrough of your model where you can show messages and reveal the model step-by-step.

Welcome to your new Insight!

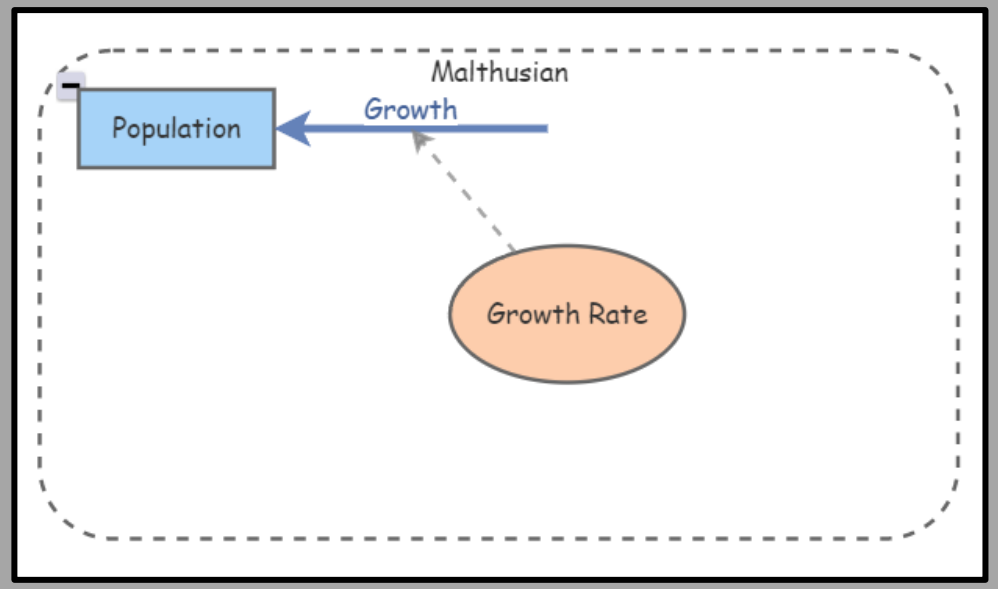
Start diagramming, modelling and simulating.

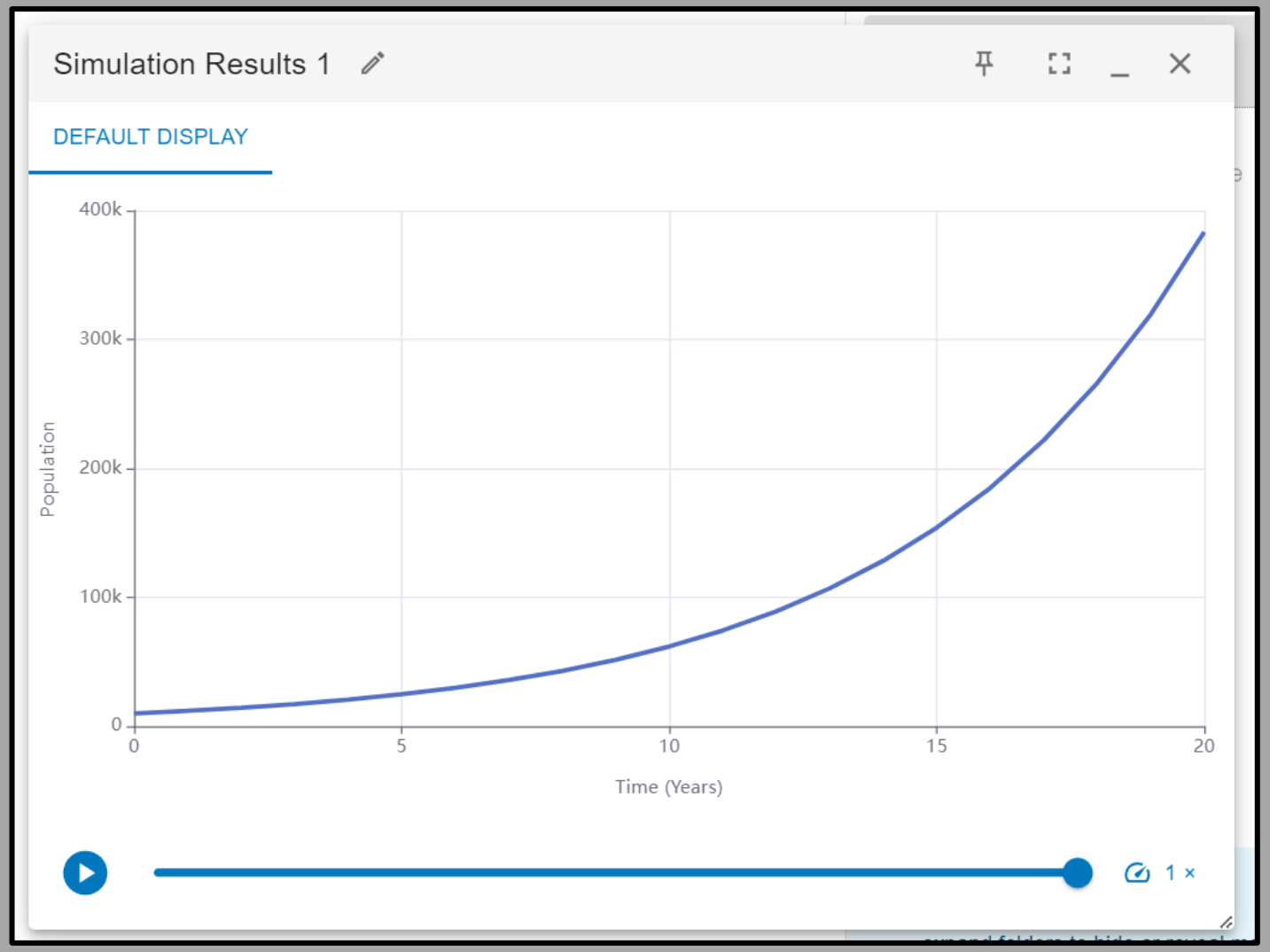
You can experiment with your model now, or take advantage of our help resources.

LEARN MORE | SAVE INSIGHT

Questions or feedback? Email support@insightmaker.com.

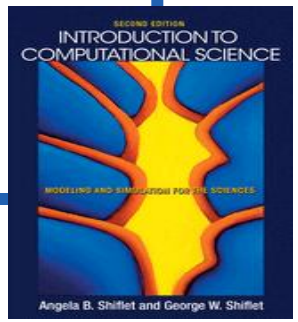
Unconstrained growth model and simulation examples using Insight Maker





System's modeling tool

- Helps to model
- Performs simulation
- What happens at one time step influences what happens at next



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Stock/Box Variable/Reservoir

- Anything that accumulates, buffer, resource
- Examples
  - Population
  - Radioactivity
  - Phosphate
  - Body fat
  - Labor

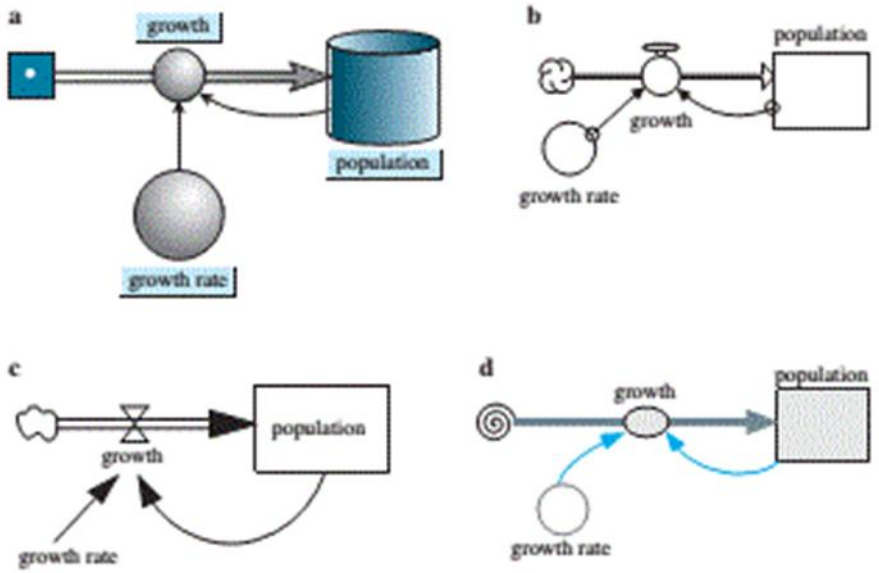
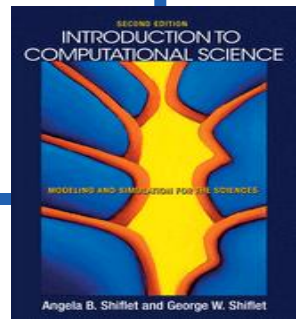


Figure 2.2.1 Diagrams of population models where growth rate is proportional to population: (a) *Berkeley Madonna*® (b) *STELLA*® (c) *Vensim PLE*® (d) Text's format

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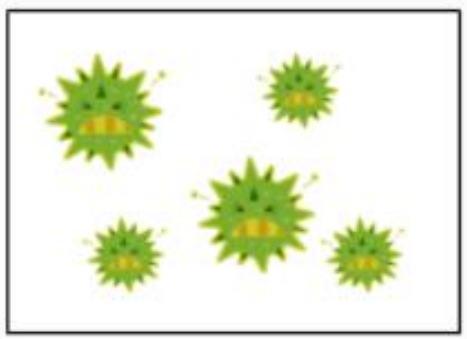
INSIGHT MAKER

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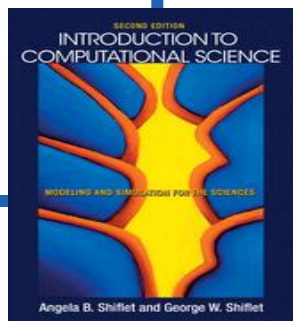
Stocks

A stock is a bucket into which a something can accumulate or be withdrawn from. What the bucket holds is specific to your model; it could be water, or money, or even deer. You can set an initial value or level for the stock that it will take on at the start of the simulation. The initial value may be a simple number or a more complex mathematical expression.





Stocks are flexible. For instance, they can hold people, water, or viruses.



Flow

- Represents activities
- Changes the magnitude of a stock
- Examples
  - Birthing, dying with population
  - Intaking & expending calories with body fat

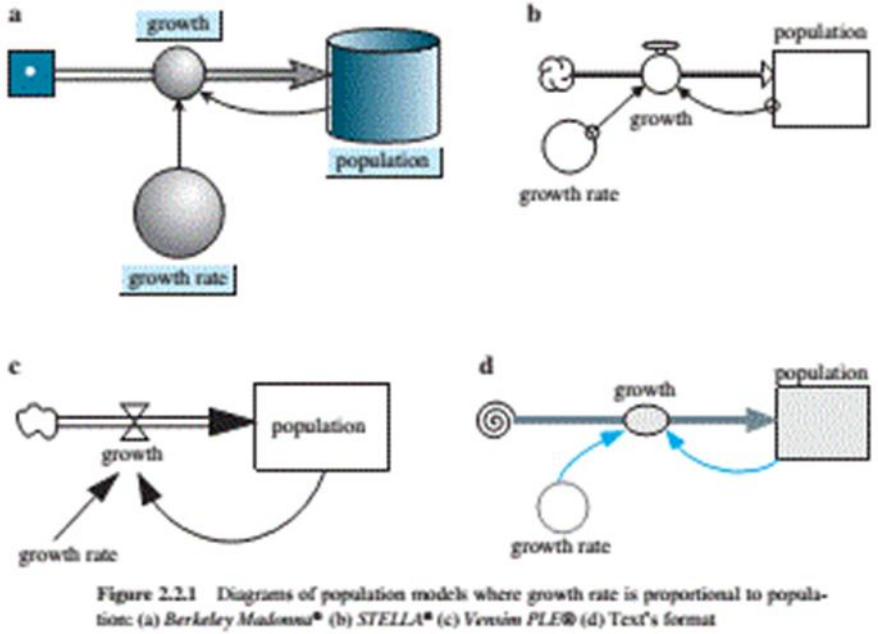
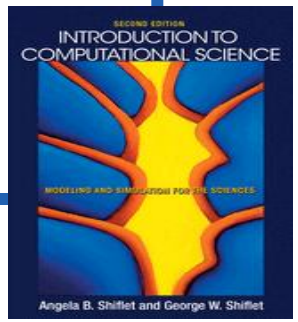


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Converter/Variable/Formula

- Contains equations that generate output for each time period
- Converts inputs into outputs
- Takes in information & transforms for use by another variable
- Examples
  - Growth rate with population & growth
  - Calories in a food

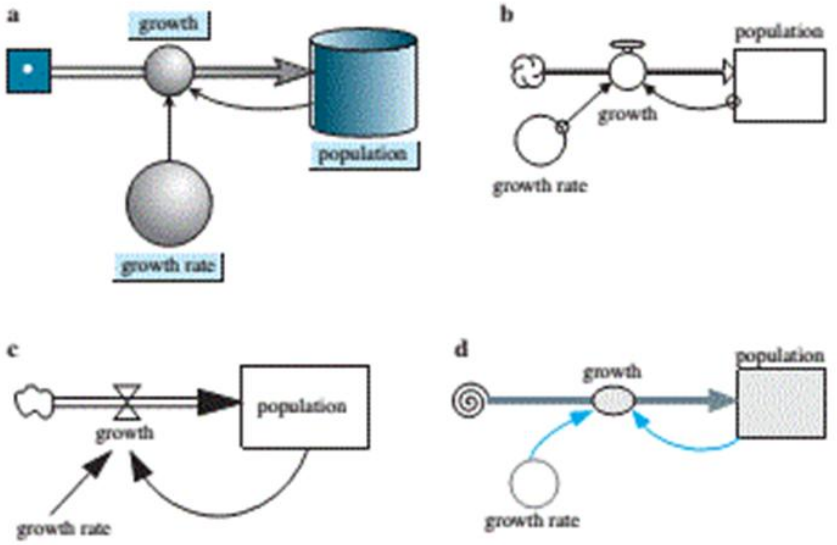


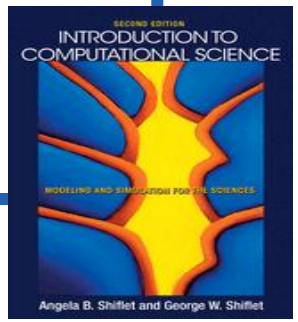
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Connector/Arrow/Arc

- Link
- Transmits information & inputs
- Regulates flows

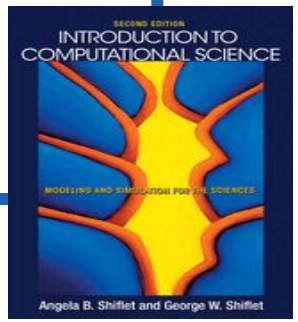
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With system dynamics tool

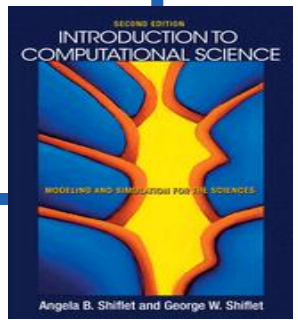
- Enter equations
- Run simulations
- Produce graphs
- Produce tables

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Algorithm for simulation of exponential growth

- initialize *simulationLength*, *population*, *growthRate*. Δt
- *numIterations* ← *simulationLength* / Δt
- for *t* going from 0 to *simulationLength* in steps of size Δt do the following:
  - *growth* ← *growthRate* * *population*
  - *population* ← *population* + *growth* * Δt
  - $t \leftarrow i * \Delta t$
  - display *t*, *growth*, and *population*



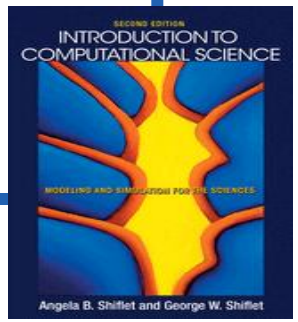
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Algorithm for simulation of exponential growth

Algorithm 1 Algorithm for simulation of unconstrained growth

initialize *simulationLength*
initialize *population*
initialize *growthRate*
initialize length of time step Δt
numIterations $\leftarrow$ *simulationLength*/Δt
for *i* going from 1 through *numIterations* do the following:
    growth $\leftarrow$ *growthRate* * *population*
    population $\leftarrow$ *population* + *growth* * Δt
    $t \leftarrow i * \Delta t$
    display *t*, *growth*, and *population*

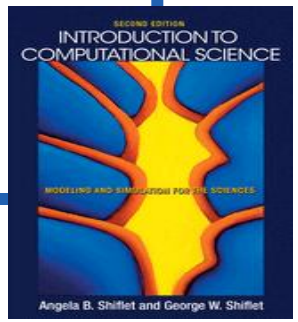
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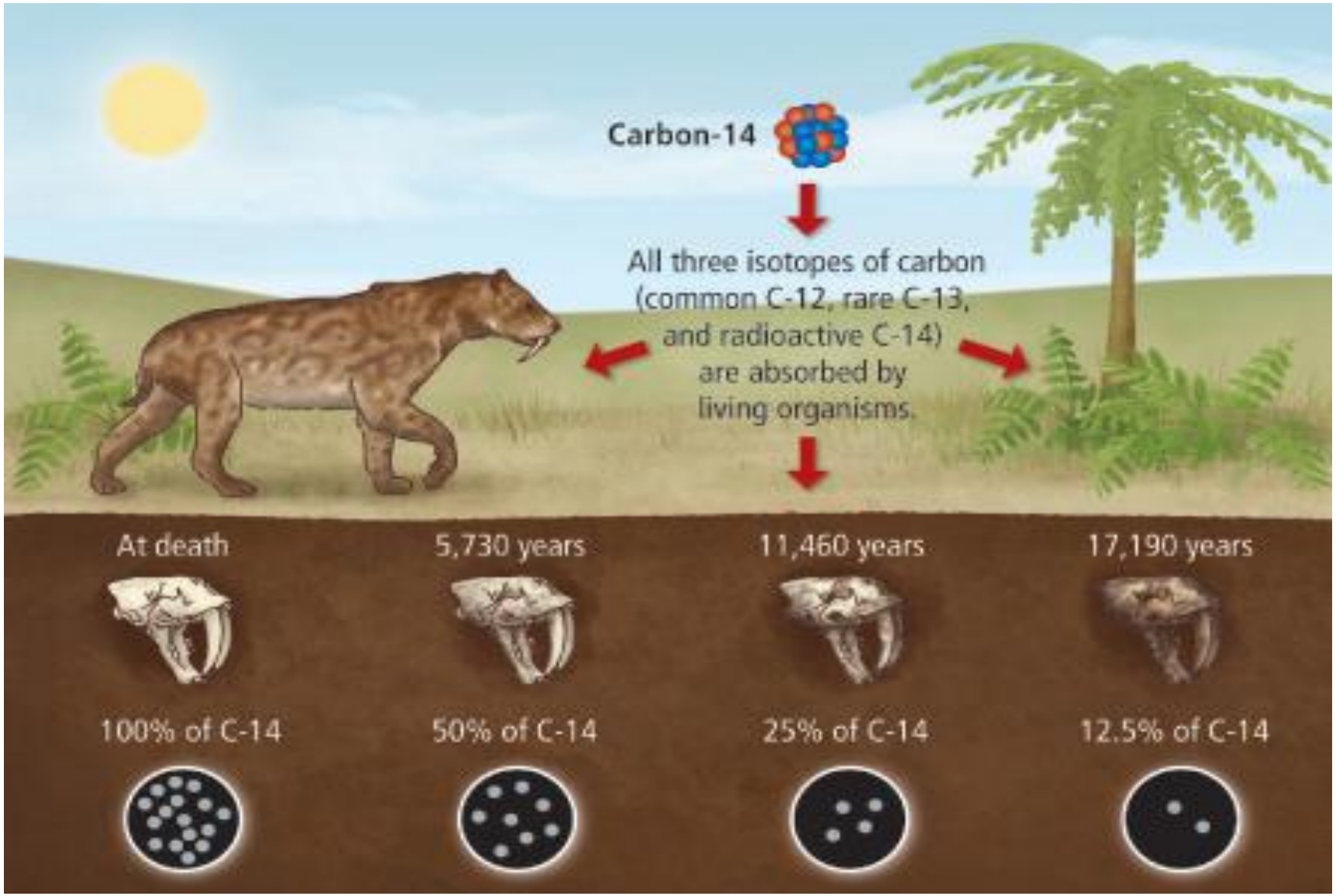


Exponential Decay

- Rate of change of mass of radioactive substance proportional to mass of substance
- Constant of proportionality negative
  - Example: Radioactive carbon-14: -0.000120968
- $dQ/dt = ?$
  - $dQ/dt = -0.000120968Q$
- $Q = Q_0 e^{-0.000120968t}$
- Carbon dating
- The **half-life** is the period of time that it takes for a radioactive substance to decay to half of its original amount Q_0, i.e. 0.5 Q_0.

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Source: https://socratic.org/questions/what-do-scientists-learn-about-fossils-by-using-carbon-14

Example: carbon dating

- What is the half-life of radioactive carbon-14 with a decay constant of -0.000120968 per year?

$$Q = 0{,}5\ Q_0$$

$$0{,}5\ Q_0 = e^{-0{,}000120968t}\ Q_0$$

$$0{,}5 = e^{-0{,}000120968t}$$

$$\ln(0{,}5) = \ln(e^{-0{,}000120968t})$$

$$-0.693 = -0{,}000120968t$$

$$t \approx 5730 \text{ tahun}$$

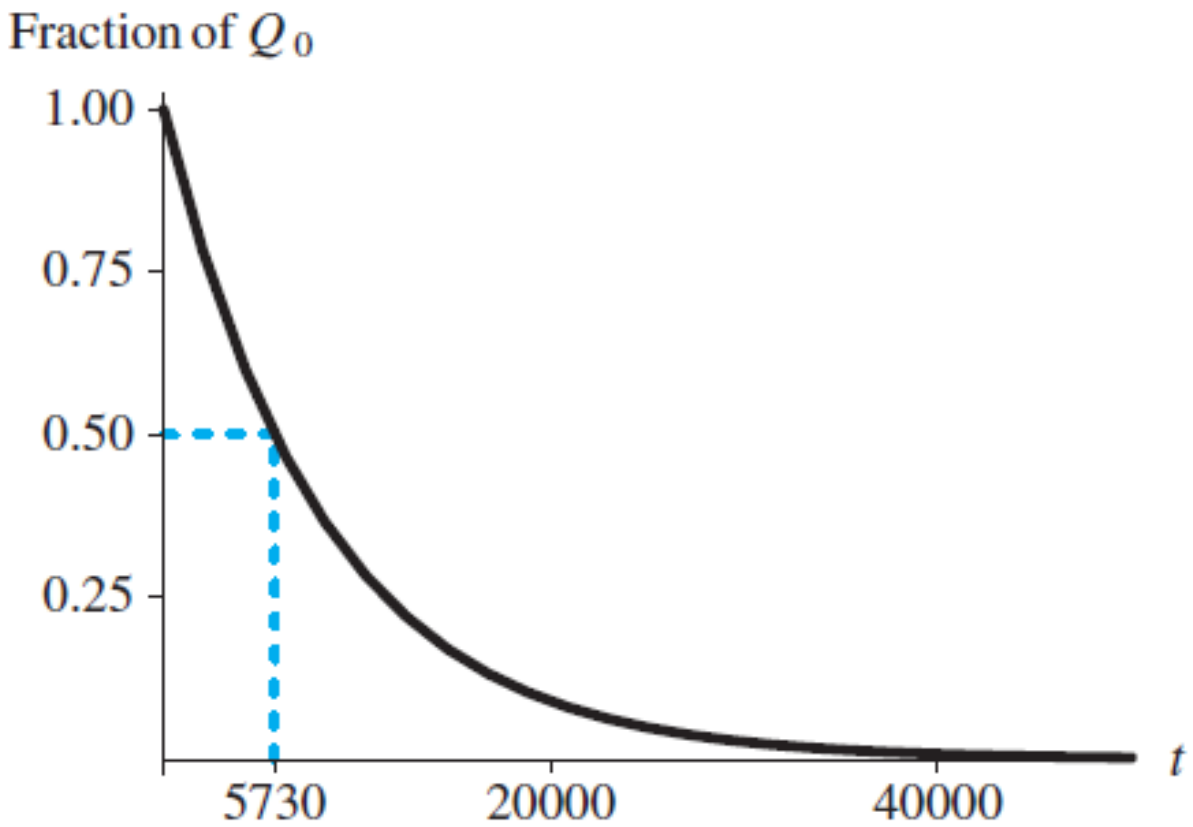
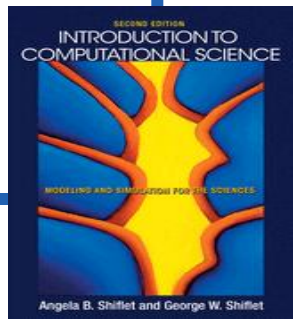


Figure 2.2.5 The half-life of radioactive carbon-14 indicated as 5730 yr

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Quick Review Questions

- Question 5
  - Radium-226 has a continuous decay rate of about 0.0427869% per year. Determine its half-life in whole years.

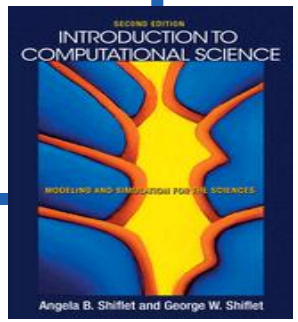
$$Q = 0.5Q_0$$

$$0.5Q_0 = Q_0 e^{-0.000427869t}$$

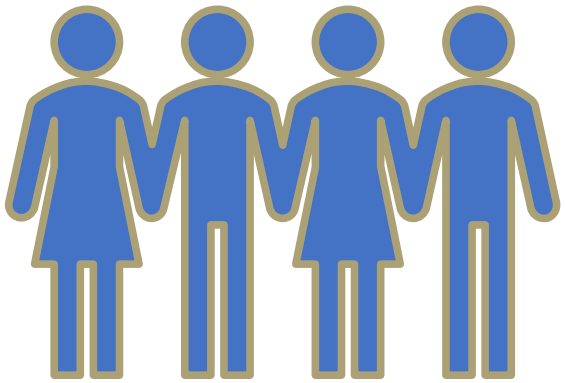
$$\ln(0.5) = \ln\left(e^{-0.000427869t}\right) = -0.000427869t$$

$$t = \frac{\ln(0.5)}{-0.000427869} \approx 1619.99 \text{ tahun}$$

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Constrained Growth and Decay



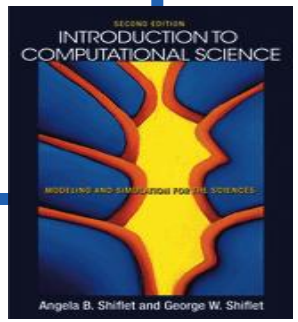
Module 2.3
Constrained Growth

Angela B. Shiflet and George W. Shiflet

Wofford College

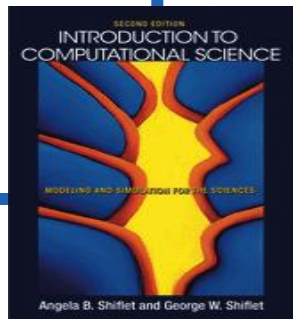
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Unconstrained Growth and Decay of population (P)

- $dP/dt = rP$
- Limitations to unconstrained growth?
- Carrying capacity (M): maximum number of organisms area can support

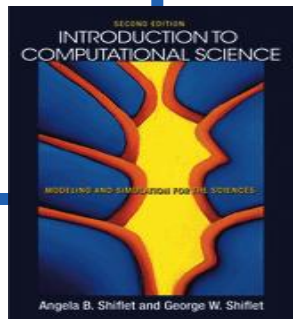


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Unconstrained Growth and Decay of population (P)

- Populations increase rapidly (such as exponential growth) at first, but they eventually encounter **resistance** from the environment – competitors, predators, limited resources, and disease.
- The maximum population size that the environment can support indefinitely is called the **carrying capacity**.

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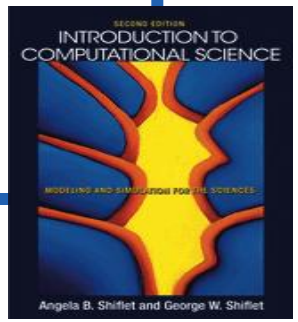
Rate of change of population

- D = number of deaths
- B = number of births
- rate of change of P = (rate of change of B) – (rate of change of D)

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}$$



- Rate of change of population:
- $\frac{dP}{dt} = rP - \frac{dD}{dt}$



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Constrained Growth

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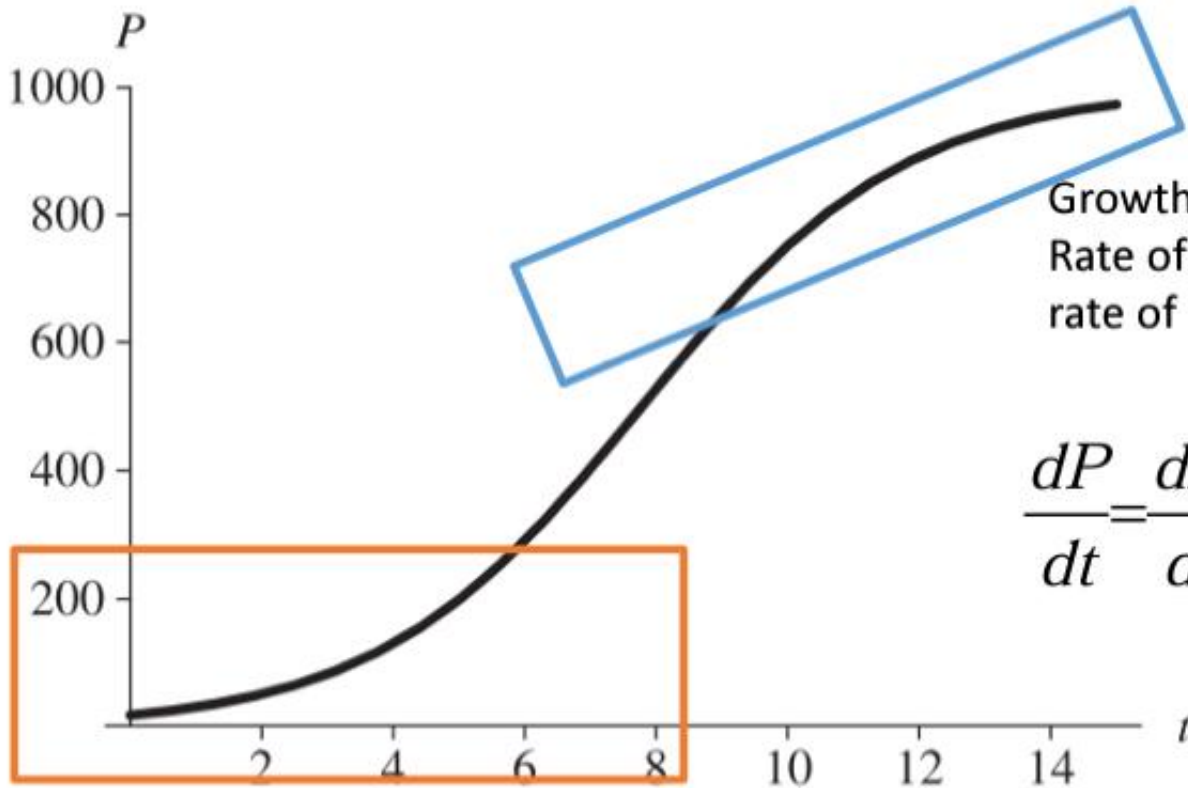
If population is less th
but close to carrying c

$$\frac{dP}{dt} \approx \frac{dB}{dt} = rP$$

If population is much less than carrying capacity

Almost unconstrained model

Rate of change of D $(\frac{dD}{dt}) \approx 0$



Growth is dampen/lessen, aln
Rate of change of D almost eq
rate of change B

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt} \approx rP - rI$$

$$\frac{dD}{dt} \approx \frac{dB}{dt}$$

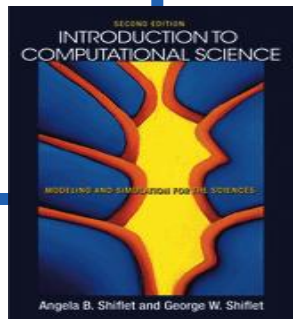
If population is much less than carrying capacity

- Almost unconstrained model
- Rate of change of D (dD/dt) ≈ 0

$$\frac{dD}{dt} \gg 0$$

$$\frac{dP}{dt} \gg \frac{dB}{dt}$$

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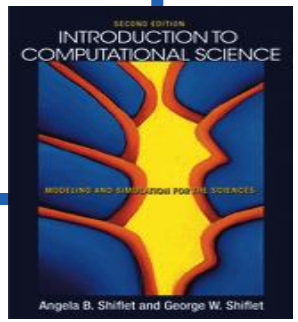


If population is less than but close to carrying capacity

- Rate of change of D larger, approaching rate of change B
- Growth is dampened, almost 0

$$\frac{dD}{dt} \gg \frac{dB}{dt} = rP$$

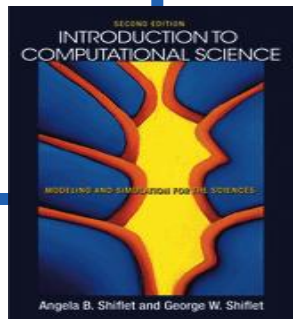
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$$\frac{dP}{dt} = rP - \frac{dD}{dt}$$

$dD/dt = death_rateP$,
multiply $death_rateP$ by something so that

- $dD/dt \approx 0$ for P much less than M
  - In this situation, f almost 0
- $dD/dt \approx dB/dt = rP$ for P less than but close to M
  - In this situation, f almost1
- What is a possible factor f?
  - One possibility is P/M



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Rate of change of population

- Rate of change of B proportional to P

$$\frac{dB}{dt} = rP$$

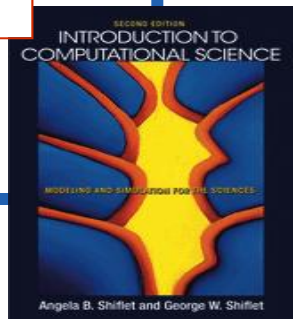
- Rate of change of D, by including carrying capacity M

$$\frac{dD}{dt} = \left(r\frac{P}{M}\right)P$$

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}$$

$$\frac{dP}{dt} = \underbrace{(rP)}_{births} - \underbrace{\left(r\frac{P}{M}\right)P}_{deaths}$$

$$\frac{dP}{dt} = r\left(1 - \frac{P}{M}\right)P$$



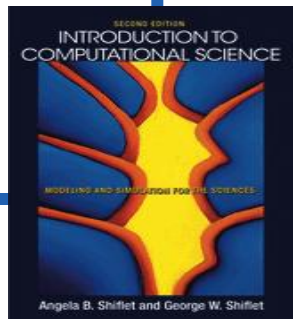
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Continuous logistic equations

$$\frac{dD}{dt} = (rP)\frac{P}{M}$$

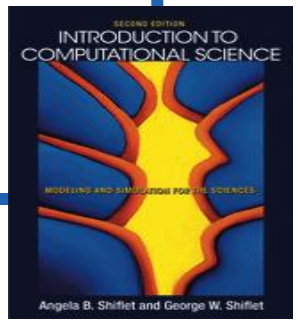
$$\frac{dP}{dt} = (rP) - (rP)\frac{P}{M} = r\left(1 - \frac{P}{M}\right)P$$

This **logistic equation** can be solved by using separation of variables.



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- The logistic equation (sometimes called the Verhulst model or logistic growth curve) is a model of population growth first published by Pierre Verhulst (1845, 1847).

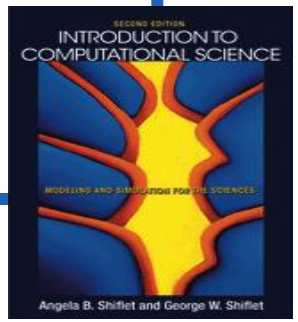


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Continuous logistic equations

If we let a = r and b = r/M, then the logistic equation can be rewritten as dP/dt = P(a – bP). Use separation of variables and partial fraction decomposition, we have $(\frac{\frac{1}{a}}{P} + \frac{\frac{b}{a}}{a-bP})\mathrm{dP} = \mathrm{dt}$, integrate and simplify, we have $P(t) = \frac{aP_0}{bP_0+(a-bP_0)e^{-at}}$ where $P_0 = P(0)$.

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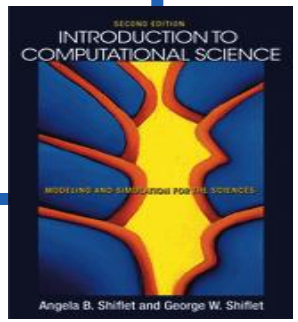
Discrete logistic equations

$$\Delta D = D(t) - D(t - \Delta t) = \left(rP(t - \Delta t)\right)\frac{P(t - \Delta t)}{M}\Delta t$$

$$\Delta P = \text{births} - \text{deaths}$$

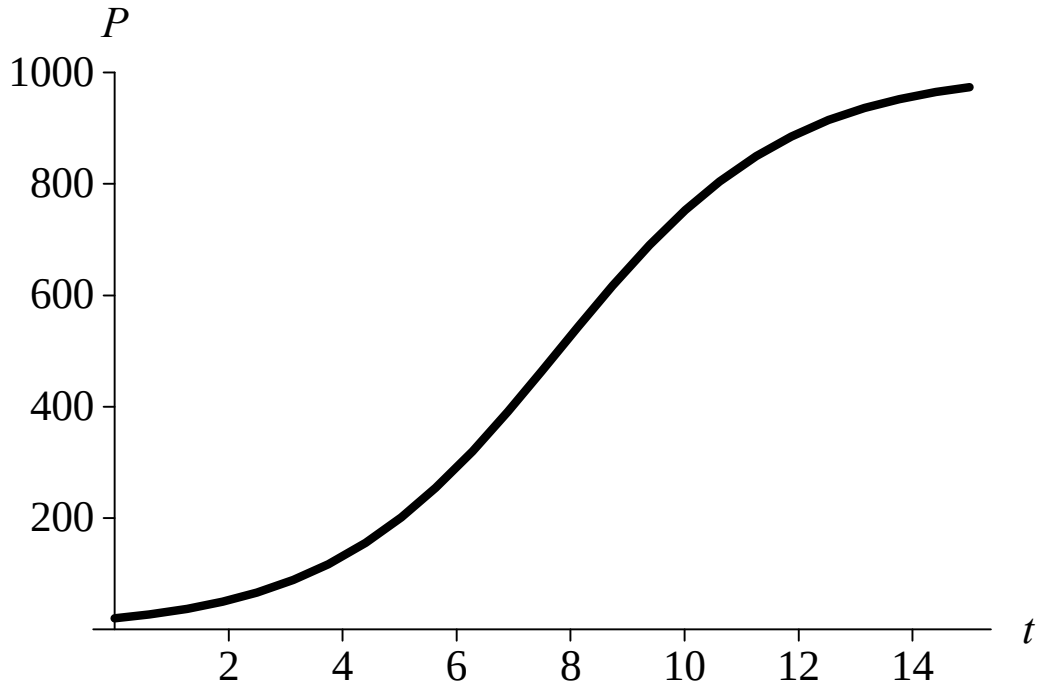
$$\Delta P = \left(rP(t - \Delta t)\right)\Delta t - \left(rP(t - \Delta t)\right)\frac{P(t - \Delta t)}{M}\Delta t$$

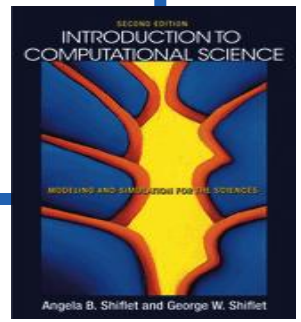
$$\Delta P = k\left(1 - \frac{P(t - \Delta t)}{M}\right)P(t - \Delta t), \text{ where } k = r\Delta t$$



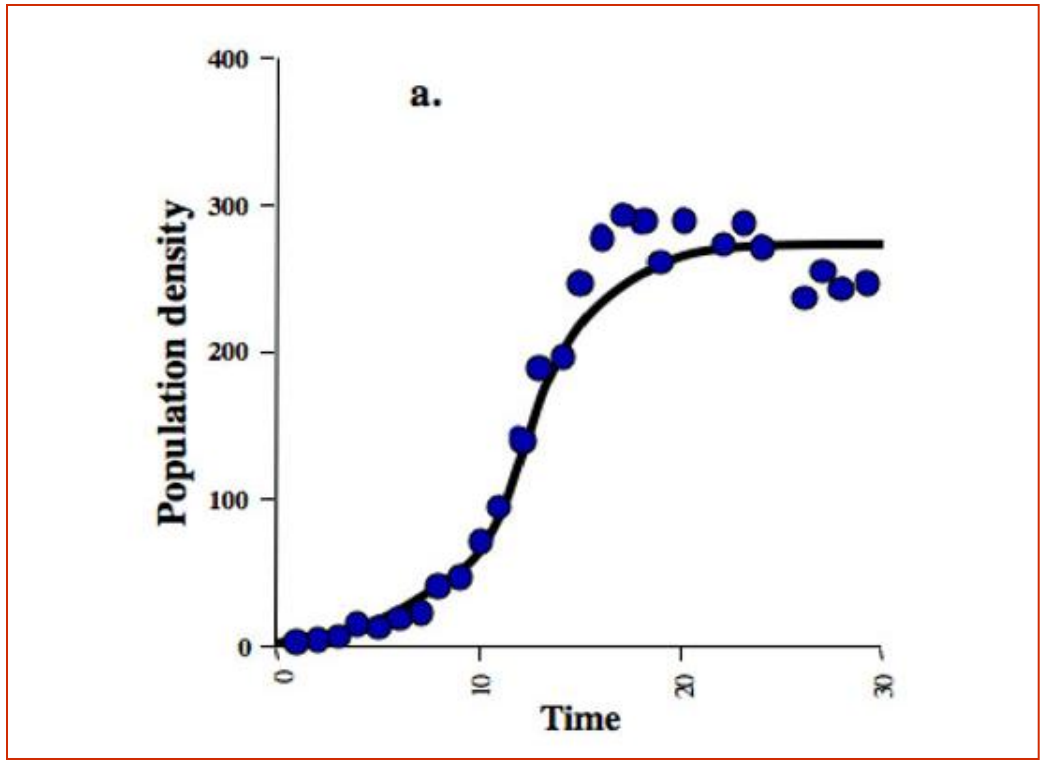
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If initial population < M, S-shaped graph





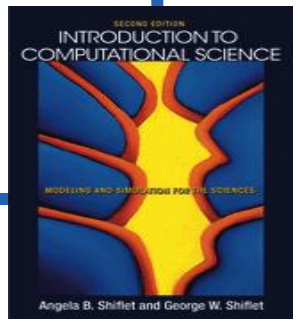
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A laboratory population of Paramecium

Note how the density first looks exponential (indeed these are the same data presented in figure 2, but over a longer time frame), but later, after the population gets to around 300 cells per cc, it levels off. This is the classic logistic pattern.

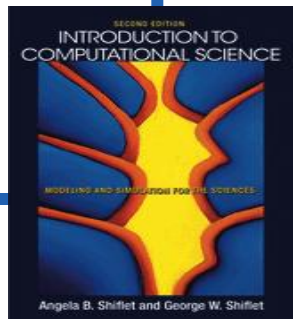
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If population greater than *M*

- What is the sign of growth?
  - Negative (P decreases in size through deaths)
- How does the rate of change of *D* compare to the rate of change of *B*?
  - Greater
- Does this situation fit the model?

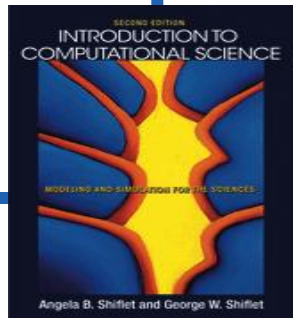
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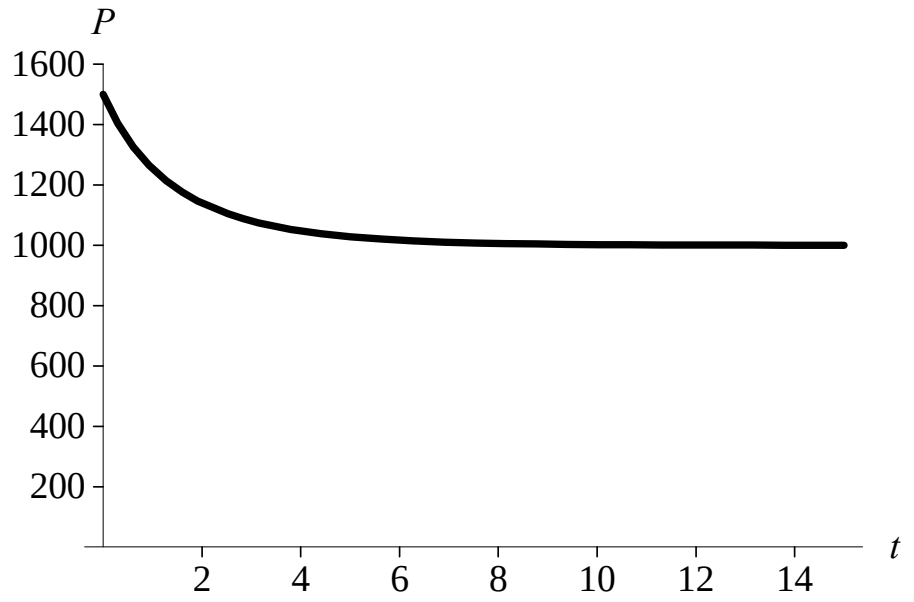
If population greater than *M*

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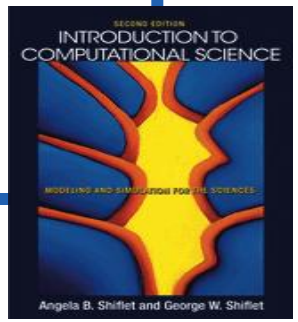
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If initial population > M



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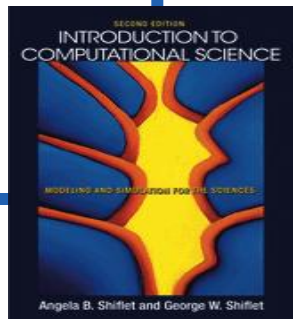


The logistic equation with carrying capacity $M = 1000$ has an interesting property.

- If the initial population is less than 1000, as in Figure 2.3.1, the population increases to a limit of 1000.
- If the initial population is greater than 1000, as in Figure 2.3.2, the population decreases to the limit of 1000.
- Moreover, if the initial population is 1000, we see that

$$\frac{P}{M} = \frac{1000}{1000} = 1 \text{ and } \frac{dP}{dt} = r(1-1)P = 0.$$

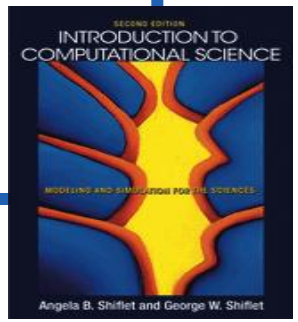
- In discrete terms, $\Delta P = 0$.
- A population **starting at the carrying capacity remains there**.
- We say that $M = 1000$ is an **equilibrium** size for the population because the population remains steady at that value or $P(t) = P(t - \Delta t) = 1000$ for all $t > 0$.



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Definitions An **equilibrium solution** for a differential equation is a solution where the derivative is always zero. An **equilibrium solution** for a difference equation is a solution where the change is always zero.

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Stability

- Equilibrium solution q is stable if there is interval (a, b) containing q, such that if initial population $P(0)$ is in that interval then
  - $P(t)$ is finite for all $t > 0$
  - $P \rightarrow q$
- $P = M$ is stable equilibrium
- $P = 0$ is unstable equilibrium

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